

SHARP TORSION STABILITY AND CERTIFIED SPECTRAL SURGERY

R. ARUN CHANDRU

ABSTRACT. We study quantitative spectral comparison under domain restriction and its use in finite spectral optimization. For arbitrary nested finite-volume open sets in $\mathbb{R}^n$, $n \geq 2$, physical torsion loss controls the critical inverse-power trace difference and the $\mathcal{S}_{1+n/2}$ power of the zero-extended resolvent difference. We determine the sharp torsion exponent in every Schatten class above the dimension threshold, with matching geometric examples, and characterize exactly the scalar spectral filters admitting uniform linear torsion control. Heat traces and regularized Fredholm determinants provide concrete applications.

A complementary low-energy comparison controls both the norm and the trace of compressed resolvent differences. In the plane it leads to explicit finite libraries for fixed-index optimization, monotone finite-spectrum objectives, and the all-domain eigenvalue-to-index infimum. The libraries have quantified cell, boundary and index budgets. Their continuum spectra are certified by compression of a spectrally clipped ambient Laplacian. We prove an explicit $H^{-1/2}$ error bound uniform over unit-cell masks, show that its decay exponent is sharp, and extend the estimate to supplied shape-regular triangular meshes. Subtracting the clipping plateau also gives finite-rank approximation of the whole resolvent in Schatten norms. Rational matrix enclosures and exact inertia complete the finite algorithms. The analysis combines classical heat-content, torsion-surgery and boundary-flux methods with sharp quantitative comparisons and explicit certification interfaces.

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Date: 2 October 2026.

2020 Mathematics Subject Classification. Primary 35P15, 47B10; Secondary 49R05, 65N25.

Key words and phrases. Dirichlet spectrum, torsional rigidity, Schatten classes, heat content, spectral surgery, shape optimization, certified eigenvalues.

1. INTRODUCTION

The Dirichlet spectrum records infinitely many responses of a domain, whereas its torsion function is the response to a single constant load. A useful quantitative comparison must explain how much spectral information that one response controls. A second question arises when domains themselves are varied: can the resulting spectral comparisons be converted into finite, certified optimization problems whose bounds do not depend on the complexity of the initial boundary?

These questions meet at the resolvent. Domain inclusion orders Dirichlet resolvents, heat kernels and torsion functions in compatible, but distinct, senses. The common structure allows an estimate of the whole spectrum and a stronger, linear estimate on a prescribed low-energy subspace. The whole-spectrum estimate identifies the exact scale of torsion stability. The low-energy estimate permits geometric surgery while retaining either an entire spectral prefix or a controlled amount of spectral rank. A separate quantitative comparison then certifies the spectra of the finite geometries produced by that surgery.

We use the actual Dirichlet form $H_0^1(D)$ on each open set D . All resolvents and functions are extended by zero when compared in a common space. Thus the boundary condition retains the information carried by sets of positive Sobolev capacity. Write

$$R_D = (-\Delta_D)^{-1}, \quad w_D = R_D 1_D, \quad T(D) = \int_D w_D \, dx,$$

and enumerate Dirichlet eigenvalues increasingly, with multiplicity. Physical torsion is denoted by T ; the negative torsion energy used in geometric minimization will be $-T/2$.

Sharp stability across the whole spectrum. Put $d = n/2$ and $r = d + 1$. For every nested pair $A \subset D$ of finite-volume open sets, the basic comparison is

$$\|R_D - R_A\|_{\mathcal{S}_r}^r \leq \sum_{j \geq 1} (\lambda_j(D)^{-r} - \lambda_j(A)^{-r}) \leq c_n [T(D) - T(A)], \quad c_n = \frac{2^{2-d}}{\pi^d \Gamma(d+1)}. \quad (1.1)$$

The empty inner set is allowed, with zero reciprocal eigenvalues. No eigenvalue simplicity or boundary regularity is required. In dimension two, (1.1) is a Hilbert–Schmidt estimate with coefficient $2/\pi$.

For fixed upper outer volume, the optimal uniform exponent of $\delta = T(D) - T(A)$ in the Schatten- p norm is

$$\alpha_n(p) = \begin{cases} 1 - \frac{n}{2p}, & n/2 < p \leq 1 + n/2, \\ \frac{1}{1 + n/2}, & 1 + n/2 \leq p \leq \infty. \end{cases} \quad (1.2)$$

We provide explicit constants and matching examples. Many small removed components determine the first branch; one shrinking removed component determines the second. The examples can keep the outer volume and first eigenvalue fixed. At $p \leq n/2$, membership itself can fail. For an arbitrary pair, the corresponding control is the overlap torsion

$$\Gamma(\Omega, \Lambda) = T(\Omega) + T(\Lambda) - 2T(\Omega \cap \Lambda).$$

Its interpretation as the least additive torsion cost of nested domain changes explains the role of the intersection in the comparison.

The critical inverse power also determines which scalar quantities of the entire ordered spectrum admit linear torsion control. Let $\psi : [0, \infty) \rightarrow \mathbb{R}$ be continuous with $\psi(0) = 0$. A finite constant independent of the nested finite-volume pair in

$$\sum_{j \geq 1} |\psi(\lambda_j(D)^{-1}) - \psi(\lambda_j(A)^{-1})| \leq C_\psi [T(D) - T(A)] \quad (1.3)$$

exists exactly when $s \mapsto \psi(s^{1/r})$ is Lipschitz on $[0, \infty)$. At a prescribed outer volume, only the reciprocal range permitted by Faber–Krahn enters. The converse follows from concentric

balls and an unchanged-component construction. This characterizes stability of scalar spectral coordinates without requiring monotonicity of the filter.

For example, the classification gives heat-trace estimates at every positive time. For the standard regularized Fredholm determinants, the planar conclusion is

$$0 \leq \log \det_2(I + zR_A) - \log \det_2(I + zR_D) \leq \frac{z^2}{\pi} [T(D) - T(A)], \quad z \geq 0.$$

In every dimension the admissible integer regularization orders can be determined exactly. Their convergence threshold and their linear torsion-stability threshold need not coincide.

Explicit certification of finite geometries. Let P be the interior of a finite union of closed unit lattice squares inside a larger square Q , and let $J : L^2(P) \rightarrow L^2(Q)$ be zero extension. The bounded operator

$$B_H = J^* \min(-\Delta_Q, H) J$$

is a finite-rank perturbation of HI . Its increasing variational levels $\ell_j(H)$, continued at the plateau H , are therefore determined by a finite Gram matrix. For $E, H \geq 1$,

$$0 \leq \lambda_j(P) - \ell_j(H) \leq 256(E+1)^{3/2} H^{-1/2}, \quad \lambda_j(P) \leq E. \quad (1.4)$$

The constant is uniform over the number of cells, holes, components and permitted point contacts. A fixed rectangular mask gives a matching lower bound proportional to $H^{-1/2}$.

The proof constructs a boundary vector field with prescribed normal component, obtains a Rellich flux estimate, and couples it to an exterior trace estimate. A squared residual identity gives quantitative large-coupling convergence. An exact harmonic-mean compression identity transfers that estimate to B_H . Sector expansions justify the boundary identity at reentrant corners and point contacts.

The same method treats masks selected from a supplied conforming triangulation of the whole ambient square. If each angle is at least $\theta > 0$ and each altitude is at least $\sigma > 0$, the right side of (1.4) is replaced by

$$4\{24 \csc(\theta/2)(1 + \csc^2 \theta) + 2\}(E + \sigma^{-2})^{3/2} H^{-1/2}.$$

This states the geometric conditioning explicitly. Rational input admits a complete arithmetic certificate, including degenerate phase differences in the triangle integrals and repeated eigenvalues.

The finite matrix has a whole-operator interpretation as well. Subtracting the plateau from the clipped inverse gives

$$F_{H,s} = (B_H + s)^{-1} - (H + s)^{-1} I.$$

This positive finite-rank operator approximates $(-\Delta_P + s)^{-1}$ in every planar Schatten class $\mathcal{S}_p$, $p > 1$, with explicit bounds. The subtraction links the finite certification hierarchy to compact resolvent stability.

Finite spectral optimization in the plane. For bounded nonempty planar open sets, define

$$\mu_k = \inf_{|\Omega|=1} \lambda_k(\Omega), \quad L = \inf_{\Omega, k \geq 1} \frac{|\Omega| \lambda_k(\Omega)}{k}. \quad (1.5)$$

The infima include disconnected sets. A library of actual polygonal domains can approximate these values even when the initial domain has arbitrary boundary complexity.

The relevant low-energy estimates are linear in torsion loss for both the norm and the trace of a compressed resolvent difference. The norm estimate preserves a full initial spectrum under torsion surgery. The trace estimate allows removed physical area to compensate for discarded spectral rank. Compact trials, regular positive torsion levels and a covering argument that charges small holes by their perimeter then produce finite grid libraries.

For $0 < \varepsilon \leq 1$, the main budgets are as follows. Here a mask with N unit cells is normalized to area one by the single dilation $P^\# = N^{-1/2} P$.

Quantity	Finite witness budgets	Guaranteed comparison
μ_k	$\lceil 125000000k^2\varepsilon^{-2} \rceil$ cells; at most k components	$\mu_k \leq \min_P \lambda_k(P^\#) \leq (1 + \varepsilon)\mu_k$
L	$\lceil 2^{46}\varepsilon^{-6} \rceil$ cells; $\lceil 302\varepsilon^{-3} \rceil$ indices; connected masks	$L \leq \min_{P,j} P \lambda_j(P)/j \leq (1 + \varepsilon)L$

The number of candidate geometries can be bounded more efficiently than by recording every occupied cell independently. Exposed boundary edges determine occupancy, giving logarithmic catalogue sizes

$$O\left(k^2\varepsilon^{-1} \log \frac{2k}{\varepsilon}\right), \quad O\left(\varepsilon^{-5} \log \frac{2}{\varepsilon}\right) \quad (1.6)$$

for the two problems. Localization supplies the required effective finite-index modulus for the second optimization.

There is also one library, chosen independently of the objective, for every computable continuous coordinatewise nondecreasing function of a fixed spectral prefix. A supplied effective modulus of continuity converts the simultaneous one-sided spectral replacement into an enclosure of the optimum and a polygonal near-minimizer.

For a mask with at most B raw unit cells and a prefix of length m , the cutoff (1.4) leads to a sufficient matrix dimension

$$O(B^4 m^3 \tau^{-2}) \quad (1.7)$$

for normalized spectral accuracy τ . Rational matrix enclosures and exact symmetric inertia give polynomial per-mask bit cost in these parameters. Consequently the scalar searches have deterministic total bit costs bounded by the exponential of the respective orders in (1.6). The estimates quantify finite exhaustive searches rather than the performance of a numerical shape optimizer.

Relation to the literature. Heat-kernel domination and heat-content comparisons provide the starting analytical tools. Van den Berg and Davies [30], and later torsion estimates such as [29], relate a domain's heat trace to its heat content. We apply the semigroup product comparison to nested differences before integrating, retaining the torsion loss. Positive trace-power inequalities and singular-value integration are classical [17]. Transformation-based Schatten and whole-spectrum stability estimates are developed in [4, 5]. The comparisons here instead use zero extension, arbitrary domain restriction and one scalar response deficit, with matching geometric tests for the full exponent range. The scalar-filter converse organizes the consequences of that endpoint.

Quantitative stability near a ball has a different geometric normalization. The resolvent and eigenfunction results of Allen, Kriventsov and Neumayer [2], and the spectral estimates of Bucur, Lamboley, Nahon and Prunier [10], are important points of comparison. The common low-energy projector methods are compatible with this literature. The surgery argument below uses a separate arbitrary-nested norm and trace comparison, proved from positivity and the low-energy projector.

Bounded simultaneous replacements and spectral optimization have substantial antecedents in Mazzoleni–Pratelli [25], Bucur [9], and Bucur–Mazzoleni [11]; the compression architecture appears already in [8]. The union structure and limiting optimal eigenvalue ratios are studied in [13, 18]. Here the quantitative assembly retains explicit cell and boundary budgets and a finite-index modulus for the unrestricted infimum. The latter concerns optimization over the shape as well as the index, whereas a domain-dependent Weyl onset, as in [19], controls high frequencies on a fixed shape.

Square-root large-coupling estimates and finer smooth spectral expansions are established in [7, 1]. The certification argument makes its constants uniform over the specified mask geometries and transfers them to the actual clipped operator. Corner analysis follows the classical sector theory [14]. Verified polygonal spectral enclosures [22] and certified finite-pixel computation [28] already provide effective spectral algorithms. The contribution of the clipping

estimate is an explicit a-priori cutoff, its sharp decay order, and its integration with the finite geometric search. The arithmetic costs are accounted for separately from the geometric enumeration.

The paper proceeds from global resolvent stability to scalar spectral quantities, then develops the complementary low-energy and geometric arguments. The final analytical section proves the clipping estimates on squares and triangles. Exact arithmetic, complexity and reproducible finite checks complete the construction.

2. WHOLE-SPECTRUM STABILITY FROM TORSION LOSS

Torsion loss controls more than a fixed spectral window. We first establish a summable comparison for the complete spectrum and for the full Dirichlet inverse. Its critical exponent is determined by the interaction of the heat-kernel dimension and the time integral defining torsion. Interpolation then gives the complete range of Schatten estimates, with matching geometric examples.

2.1. Dirichlet forms, zero extension, and normalization. Throughout this section $n \geq 2$ and every domain is an open subset of $\mathbb{R}^n$ of finite measure. The empty set is allowed. For a nonempty domain D , the space $H_0^1(D)$ means the closure of $C_c^\infty(D)$ in $H^1(\mathbb{R}^n)$ after zero extension. Thus constraints on sets of positive Sobolev capacity are retained, including measure-zero slits. The Dirichlet Laplacian is associated with the form

$$q_D(u, v) = \int_{\mathbb{R}^n} \nabla u \cdot \overline{\nabla v} \, dx, \quad u, v \in H_0^1(D).$$

We denote its inverse, followed and preceded by zero extension and restriction, by R_D . Operators associated with several domains are always compared on the same ambient $L^2(\mathbb{R}^n)$, or equivalently on the L^2 space of their union.

Finite measure suffices for a positive Poincaré constant and compact resolvent. Indeed, the Sobolev inequality and Hölder's inequality give the Poincaré estimate for $n \geq 3$; in dimension two one can use the L^4 Gagliardo–Nirenberg inequality. A bounded sequence in the form norm has a locally L^2 -convergent subsequence by Rellich compactness. For $n \geq 3$, its squared L^2 norm on $D \setminus B_L$ is bounded by a constant times $|D \setminus B_L|^{2/n}$; in dimension two the L^4 estimate gives the power $1/2$. These tails tend uniformly to zero as $L \rightarrow \infty$, proving global compactness. In particular, the Dirichlet eigenvalues can be listed with multiplicity as

$$0 < \lambda_1(D) \leq \lambda_2(D) \leq \cdots \rightarrow \infty, \quad \mu_j(D) = \lambda_j(D)^{-1}.$$

The sequence $\mu_j(D)$ lists the positive eigenvalues of R_D in decreasing order. We put $R_\emptyset = 0$ and $\mu_j(\emptyset) = 0$ for all j .

Let $w_D = R_D 1_D$ and use physical torsional rigidity

$$T(D) = \int_D w_D \, dx = \int_{\mathbb{R}^n} |\nabla w_D|^2 \, dx, \quad T(\emptyset) = 0. \quad (2.1)$$

Equivalently, $-T(D)/2$ is the minimum on $H_0^1(D)$ of $u \mapsto \frac{1}{2} \int |\nabla u|^2 - \int u$. For $A \subset D$, write

$$\delta = T(D) - T(A), \quad S = R_D - R_A. \quad (2.2)$$

The torsion-energy loss is $\delta/2$. The variational formula

$$\langle f, R_D f \rangle = \sup_{u \in H_0^1(D)} \left(2 \operatorname{Re} \langle f, u \rangle - \int |\nabla u|^2 \right) \quad (2.3)$$

implies $0 \leq R_A \leq R_D$, hence $0 \leq S \leq R_D$. The comparison principle also gives $0 \leq w_A \leq w_D$ almost everywhere. Testing the two torsion equations with w_A gives

$$\delta = \int_{\mathbb{R}^n} |\nabla (w_D - w_A)|^2 \, dx. \quad (2.4)$$

For a compact operator K , write $s_j(K)$ for its singular values, $\|K\|_{\mathcal{S}_p} = (\sum_j s_j(K)^p)^{1/p}$ for $p > 0$, and $\|K\|_{\mathcal{S}_\infty} = \|K\|_{L^2 \rightarrow L^2}$. When $p < 1$, this notation denotes the usual Schatten quasi-norm. We use

$$d = \frac{n}{2}, \quad r = d + 1, \quad a_n = \left(\frac{e}{4\pi d}\right)^d, \quad c_n = \frac{2^{2-d}}{\pi^d \Gamma(d+1)}. \quad (2.5)$$

Here $\Gamma(s)$ is Euler's gamma function; below, $\Gamma(D, E)$ with two domain arguments denotes an overlap defect.

2.2. The critical Schatten and spectral-trace comparison. For every nested pair $A \subset D$ as above,

$$\|R_D - R_A\|_{\mathcal{S}_r}^r \leq \sum_{j \geq 1} (\mu_j(D)^r - \mu_j(A)^r) \leq c_n(T(D) - T(A)). \quad (2.6)$$

All terms in the middle sum are nonnegative. In particular, for planar domains this becomes

$$\|R_D - R_A\|_{\text{HS}}^2 \leq \sum_{j \geq 1} (\lambda_j(D)^{-2} - \lambda_j(A)^{-2}) \leq \frac{2}{\pi} \delta. \quad (2.7)$$

We derive the spectral-trace inequality by applying the classical heat-content comparison mechanism of [30, Lemma 2.6] to a nested difference. Let $p_D(t, x, y)$ be the zero-extended Dirichlet heat kernel, and put

$$Z_D(t) = \sum_j e^{-t\lambda_j(D)}, \quad Q_D(t) = \iint_{\mathbb{R}^n \times \mathbb{R}^n} p_D(t, x, y) \, dx \, dy.$$

The standard Dirichlet domination estimates [15] give, almost everywhere,

$$0 \leq p_A(t, x, y) \leq p_D(t, x, y) \leq (4\pi t)^{-d} \exp\left(-\frac{|x-y|^2}{4t}\right). \quad (2.8)$$

In particular $Z_D(t) \leq |D|(4\pi t)^{-d}$. At large times, the spectral bottom gives $Z_D(t) \leq e^{-\lambda_1(D)t/2} Z_D(t/2)$. Consequently R_D^p is trace class for every $p > d$, and

$$\text{Tr } R_D^p = \frac{1}{\Gamma(p)} \int_0^\infty t^{p-1} Z_D(t) \, dt, \quad T(D) = \int_0^\infty Q_D(t) \, dt. \quad (2.9)$$

The second identity follows by integrating the heat semigroup against $1_D \in L^2(D)$; both integrals are finite.

Symmetry and the semigroup law give

$$\begin{aligned} Z_D(2t) - Z_A(2t) &= \iint (p_D(t, x, y)^2 - p_A(t, x, y)^2) \, dx \, dy \\ &\leq 2(4\pi t)^{-d} (Q_D(t) - Q_A(t)). \end{aligned} \quad (2.10)$$

The subtraction occurs before estimation: its integrand factors as $(p_D - p_A)(p_D + p_A)$ with both factors nonnegative. Thus this step uses pointwise kernel order, not positive-semidefinite order of the heat-semigroup difference. Substituting $t = 2s$ in (2.9), taking $p = r = d + 1$, and then using (2.10) yields

$$\begin{aligned} \text{Tr } R_D^r - \text{Tr } R_A^r &\leq \frac{2^{r+1}(4\pi)^{-d}}{\Gamma(r)} \int_0^\infty (Q_D(s) - Q_A(s)) \, ds \\ &= c_n \delta. \end{aligned} \quad (2.11)$$

The exponent of s remaining in the integral is exactly zero. For comparison, the same one-domain computation with $A = \emptyset$ omits the factor two and gives $\text{Tr } R_D^r \leq (c_n/2)T(D)$.

To pass from the trace difference to the complete operator, we use the positive trace-power inequality

$$\text{Tr}(X + Y)^p \geq \text{Tr } X^p + \text{Tr } Y^p, \quad X, Y \geq 0, \quad p \geq 1, \quad (2.12)$$

as in [17, Proposition 4.6(ii)]. Here is an argument covering all exponents needed below. For positive matrices, differentiation under the trace gives

$$\mathrm{Tr}(X + Y)^p - \mathrm{Tr} X^p = p \int_0^1 \mathrm{Tr}(Y(X + tY)^{p-1}) dt.$$

If $p \geq 2$, choose an orthonormal eigenbasis v_i of Y , with eigenvalues y_i . Scalar spectral Jensen gives

$$\langle v_i, (X + tY)^{p-1} v_i \rangle \geq \langle v_i, (X + tY) v_i \rangle^{p-1} \geq (ty_i)^{p-1}.$$

If $1 < p < 2$, put $q = p - 1$. The representation

$$B^q = \frac{\sin(\pi q)}{\pi} \int_0^\infty s^{q-1} B(B + sI)^{-1} ds$$

and inverse order show $(X + tY)^q \geq (tY)^q$. In either case the integral is at least $\mathrm{Tr} Y^p$; the case $p = 1$ is equality. Finite-rank compressions tending strongly to the identity converge in $\mathcal{S}_p$ on compact operators in that ideal, so the matrix inequality passes to this setting. Since $0 \leq S \leq R_D$, the min-max principle already gives $S \in \mathcal{S}_r$. Apply (2.12) to $R_D = R_A + S$ and combine with (2.11) to obtain (2.6).

The same argument also treats ordered killing on a fixed finite-volume ambient domain. For nonnegative locally integrable potentials $V \leq W$, use the closed forms

$$q_V(u) = \int_D |\nabla u|^2 dx + \int_D V|u|^2 dx, \quad u \in H_0^1(D), \quad \int_D V|u|^2 < \infty.$$

Their inverses R_V, R_W satisfy the requisite form and heat-kernel orders. With $T_V = \langle 1_D, R_V 1_D \rangle$,

$$\|R_V - R_W\|_{\mathcal{S}_r}^r \leq \mathrm{Tr} R_V^r - \mathrm{Tr} R_W^r \leq c_n(T_V - T_W). \quad (2.13)$$

Actual hard restrictions are imposed through $H_0^1(A)$ itself. This convention distinguishes a Dirichlet restriction from mere almost-everywhere vanishing on its complement.

2.3. The full range of exponents. For $d < p \leq r$, set

$$C_{n,p} = \frac{p}{p-d} r^{-(p-d)} a_n^{r-p} c_n^{p-d}. \quad (2.14)$$

The nested comparison throughout this range is

$$\|R_D - R_A\|_{\mathcal{S}_p}^p \leq \mathrm{Tr} R_D^p - \mathrm{Tr} R_A^p \leq C_{n,p} |D|^{r-p} \delta^{p-d}. \quad (2.15)$$

For $r \leq p \leq \infty$ one has

$$\|R_D - R_A\|_{\mathcal{S}_p} \leq (c_n \delta)^{1/r}. \quad (2.16)$$

At $p = r$, (2.14) gives $C_{n,r} = c_n$.

To prove the lower range with the stated constant, define $N_K(t) = \#\{j : s_j(K) > t\}$. For $u, t > 0$,

$$N_{R_D}(t) \leq e^{u/t} Z_D(u) \leq |D| e^{u/t} (4\pi u)^{-d}.$$

The choice $u = dt$ gives

$$N_{R_D}(t) \leq a_n |D| t^{-d}. \quad (2.17)$$

Both the count N_S and the nonnegative difference $N_{R_D} - N_{R_A}$ admit this bound. Each has critical moment at most $c_n \delta$ by (2.6).

We record the elementary moment calculation explicitly. Suppose $F \geq 0$ is measurable,

$$F(t) \leq M t^{-d}, \quad r \int_0^\infty t^{r-1} F(t) dt \leq K, \quad r = d + 1.$$

For $d < p < r$, put $\beta = p - d \in (0, 1)$. The p -moment is finite by splitting the integral at any positive threshold. For $b > 0$, subtract the weight $b^{p-r}t^{r-1}$ from t^{p-1} , discard the nonpositive part above b , and use the two assumptions to obtain

$$\begin{aligned} p \int_0^\infty t^{p-1} F(t) dt &\leq pM \int_0^b (t^{p-d-1} - b^{p-r}t^{r-d-1}) dt + \frac{p}{r} K b^{p-r} \\ &= \frac{pM(1-\beta)}{\beta} b^\beta + \frac{p}{r} K b^{\beta-1}. \end{aligned} \quad (2.18)$$

For $M, K > 0$ the right side is minimized at $b = K/(rM)$, giving

$$p \int_0^\infty t^{p-1} F(t) dt \leq \frac{p}{p-d} r^{-(p-d)} M^{r-p} K^{p-d}. \quad (2.19)$$

If $M = 0$ or $K = 0$, the conclusion follows from $F = 0$ almost everywhere. At $p = r$ the formula is the assumed moment bound. No monotonicity of F was used.

The layer-cake identity $\text{Tr}|K|^p = p \int_0^\infty t^{p-1} N_K(t) dt$ and (2.19), with $M = a_n|D|$ and $K = c_n\delta$, give the upper bound in (2.15) by taking $F = N_{R_D} - N_{R_A}$. Its first inequality is (2.12). Taking $F = N_S$ gives the same operator bound directly. Finally, monotonicity of sequence norms $\|K\|_{\mathcal{S}_p} \leq \|K\|_{\mathcal{S}_r}$ for $p \geq r$ proves (2.16). This argument is a concrete weak/strong moment interpolation in the standard singular-value framework [17].

In the plane, (2.15) specializes to

$$\|R_D - R_A\|_{\mathcal{S}_p}^p \leq \text{Tr } R_D^p - \text{Tr } R_A^p \leq \frac{p}{\pi(p-1)} \left(\frac{e}{4}\right)^{2-p} |D|^{2-p} \delta^{p-1}, \quad 1 < p \leq 2. \quad (2.20)$$

For $p \geq 2$, the norm is at most $\sqrt{2\delta/\pi}$.

2.4. Arbitrary pairs and ordered spectral coordinates. Let D, E be finite-volume open sets and define

$$I = D \cap E, \quad V = |D| + |E|, \quad \Gamma(D, E) = T(D) + T(E) - 2T(I). \quad (2.21)$$

The arbitrary-pair bounds retain exactly the constants above:

$$\|R_D - R_E\|_{\mathcal{S}_p}^p \leq C_{n,p} V^{r-p} \Gamma(D, E)^{p-d}, \quad d < p \leq r, \quad (2.22)$$

$$\|R_D - R_E\|_{\mathcal{S}_p} \leq (c_n \Gamma(D, E))^{1/r}, \quad r \leq p \leq \infty. \quad (2.23)$$

There is also the summable coordinate estimate

$$\sum_{j \geq 1} |\mu_j(D)^p - \mu_j(E)^p| \leq C_{n,p} V^{r-p} \Gamma(D, E)^{p-d}, \quad d < p \leq r. \quad (2.24)$$

In particular the right side of (2.24) also bounds $\sum_j |\mu_j(D) - \mu_j(E)|^p$.

For the operator estimates put $U = R_D - R_I \geq 0$ and $W = R_E - R_I \geq 0$. Since $U - W \leq U$ and $W - U \leq W$, the min-max principle bounds the positive eigenvalues of $U - W$ by those of U , and the absolute values of its negative eigenvalues by those of W . For every $p > 0$,

$$\text{Tr}|U - W|^p \leq \text{Tr } U^p + \text{Tr } W^p. \quad (2.25)$$

At $p = r$, the two nested estimates therefore give $\text{Tr}|R_D - R_E|^r \leq c_n \Gamma(D, E)$. Moreover $U \leq R_D$ and $W \leq R_E$, so

$$N_{R_D - R_E}(t) \leq N_{R_D}(t) + N_{R_E}(t) \leq a_n V t^{-d}.$$

Apply (2.19) once to this combined counting function to prove (2.22); ideal inclusion gives (2.23).

For the coordinate estimate use instead

$$F(t) = N_{R_D}(t) - N_{R_I}(t) + N_{R_E}(t) - N_{R_I}(t).$$

This is nonnegative, at most $a_n V t^{-d}$, and has critical moment at most $c_n \Gamma(D, E)$. Its p -moment bounds the left side of (2.24), because $\mu_j(I)^p \leq \min(\mu_j(D)^p, \mu_j(E)^p)$ at each ordered index.

Formula (2.19) applies even though F need not be decreasing. The last assertion follows from $|x - y|^p \leq |x^p - y^p|$ for $x, y \geq 0$ and $p \geq 1$.

The critical coordinate estimate can also be expressed as

$$\int_0^\infty t^{r-1} \sum_{j \geq 1} |e^{-t\lambda_j(D)} - e^{-t\lambda_j(E)}| dt \leq \Gamma(r) c_n \Gamma(D, E). \quad (2.26)$$

An empty-domain heat coordinate is zero. For each nonempty coordinate pair the sign of the exponential difference is independent of t ; the Mellin identity and Tonelli thus identify the integral with $\Gamma(r) \sum_j |\mu_j(D)^r - \mu_j(E)^r|$.

2.5. Sharpness with fixed volume and fixed ground state. Under an upper bound on outer volume, the largest uniform power of δ in the Schatten norm is

$$\alpha(p) = \begin{cases} 1 - d/p, & d < p \leq r, \\ 1/r, & r \leq p \leq \infty. \end{cases} \quad (2.27)$$

For $0 < p \leq d$, finiteness itself can fail. We prove these assertions using smooth disconnected domains in one fixed bounded box, with exactly fixed outer volume and the same fixed first eigenvalue before and after deletion.

Write B_1 for the unit ball, $\omega_n = |B_1|$, $t_n = T(B_1) = \omega_n/[n(n+2)]$, and $z_{n,p} = \text{Tr } R_{B_1}^p < \infty$ for $p > d$. For an integer $k \geq 1$, set $N = k^n$ and $\rho_N = 1/(4k)$. Retain a unit ball A , and place N balls of radius ρ_N at the centers of a $k \times \cdots \times k$ grid in a unit cube separated from A . Their diameters are half the grid spacing, so their closures are disjoint. Let D_N be their union with A . Then

$$|D_N| = \omega_n(1 + 4^{-n}), \quad \lambda_1(D_N) = \lambda_1(A) = \lambda_1(B_1).$$

The direct-sum decomposition of the Dirichlet form gives exactly

$$\delta_N = t_n N \rho_N^{n+2}, \quad \|R_{D_N} - R_A\|_{\mathcal{S}_p}^p = z_{n,p} N \rho_N^{2p}. \quad (2.28)$$

Since δ_N is a positive constant times $N^{-1/d}$, (2.28) implies $\|R_{D_N} - R_A\|_{\mathcal{S}_p} \asymp \delta_N^{1-d/p}$. For $d < p < r$ this excludes every larger power of δ . The trace difference $\text{Tr } R_{D_N}^p - \text{Tr } R_A^p$ equals the same small-ball sum, so the exponent $p - d$ in (2.15) is sharp as well.

For $r \leq p \leq \infty$, fix $0 < \rho < 1/8$ and set

$$b(\rho) = (4^{-n} - \rho^n)^{1/n}.$$

At three fixed well-separated centers, form $A_\rho = B_1 \dot{\cup} B_{b(\rho)}$ and $D_\rho = A_\rho \dot{\cup} B_\rho$. Again $|D_\rho| = \omega_n(1 + 4^{-n})$ and both first eigenvalues equal $\lambda_1(B_1)$. On the deleted component,

$$\|R_{D_\rho} - R_{A_\rho}\|_{\mathcal{S}_p} = \rho^2 \|R_{B_1}\|_{\mathcal{S}_p}, \quad \delta_\rho = t_n \rho^{n+2}. \quad (2.29)$$

The operator-norm interpretation is used at $p = \infty$. Thus every power greater than $1/r = 2/(n+2)$ fails under the same fixed-scale constraints.

For completeness, the inverse of a ball is outside $\mathcal{S}_p$ when $p \leq d$ without needing its exact Weyl constant. A ball contains a cube, whose Dirichlet eigenvalues are a fixed positive multiple of $k_1^2 + \cdots + k_n^2$, $k_i \geq 1$. The dyadic block $N < k_i \leq 2N$ contributes at least a positive constant times N^{n-2p} to the inverse-power sum. Disjoint dyadic blocks therefore give divergence for $2p \leq n$. Domain monotonicity transfers divergence to the ball. Deleting one ball component leaves its inverse as a direct summand of $R_D - R_A$, proving the asserted failure of finiteness.

These examples establish exponent sharpness within the stated arbitrary-open-set class. The constants c_n and $C_{n,p}$ are explicit bounds rather than asserted optimal numerical constants. Their scaling is consistent: under dilation by L , the right side of (2.15) scales by $L^{n(r-p)+(n+2)(p-d)} = L^{2p}$, exactly as the left side.

Figure 1 displays the planar exponent range.

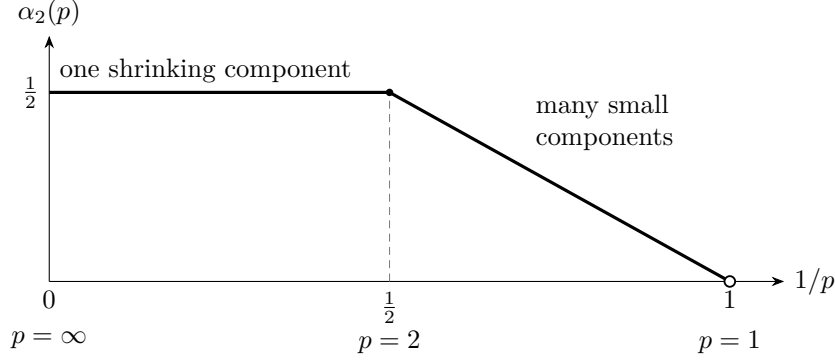


FIGURE 1. The sharp planar torsion exponent, plotted against $1/p$. The horizontal branch is forced by a single shrinking deleted component; the sloping branch by many small deleted components. The bounds hold for all finite-area nested open sets with uniformly bounded outer area. Matching examples keep the outer area and ground-state eigenvalue fixed. The endpoint $p = 1$ is excluded because trace-class membership can fail.

3. OVERLAP TORSION AND THE COST OF NESTED COMPARISONS

The quantity $\Gamma(D, E)$ in (2.21) has an exact interpretation in terms of nested domain changes. The underlying torsional supermodularity is classical [24], as is the associated valuation-metric construction [21]; the following variational proof specifies it for the actual Dirichlet form spaces used here.

For nonnegative $u \in H_0^1(D)$ and $v \in H_0^1(E)$, the functions $\min(u, v)$ and $\max(u, v)$ belong respectively to $H_0^1(D \cap E)$ and $H_0^1(D \cup E)$. This follows from the quasi-continuous characterization of H_0^1 by vanishing quasi-everywhere off the open set, or by approximation and the Sobolev lattice identities. For $J(u) = \frac{1}{2} \int |\nabla u|^2 - \int u$ those identities give

$$J(\min(u, v)) + J(\max(u, v)) = J(u) + J(v).$$

Take $u = w_D$, $v = w_E$ and minimize on the intersection and union to obtain

$$T(D \cup E) + T(D \cap E) \geq T(D) + T(E). \quad (3.1)$$

Applying this inequality to $D \cap F$ and $E \cap F$, and then using monotonicity, gives

$$T(D \cap F) + T(E \cap F) \leq T((D \cup E) \cap F) + T(D \cap E \cap F) \leq T(F) + T(D \cap E).$$

Rearrangement proves the triangle inequality

$$\Gamma(D, E) \leq \Gamma(D, F) + \Gamma(F, E). \quad (3.2)$$

Thus Γ is a pseudometric on finite-volume open sets.

For a finite chain $D = D_0, D_1, \dots, D_N = E$ in which every consecutive pair is nested in either direction, assign cost

$$\mathcal{C}(D_0, \dots, D_N) = \sum_{i=1}^N |T(D_i) - T(D_{i-1})|.$$

On a nested pair, Γ is precisely this absolute torsion difference. Repeated application of (3.2) therefore bounds every chain cost from below by $\Gamma(D, E)$. The chain $D, D \cap E, E$ attains that bound. Consequently

$$\Gamma(D, E) = \min_{\substack{\text{finite nested chains} \\ D \rightsquigarrow E}} \mathcal{C}(D_0, \dots, D_N). \quad (3.3)$$

The chain space includes the empty domain with torsion zero, and intermediate volumes are allowed to vary.

The ordinary torsion-function distance $\gamma(D, E) = \int_{\mathbb{R}^n} |w_D - w_E| dx$ obeys

$$\gamma(D, E) \leq \Gamma(D, E), \quad \Gamma(D, E) - \gamma(D, E) = 2 \int_{\mathbb{R}^n} (\min(w_D, w_E) - w_{D \cap E}) dx. \quad (3.4)$$

Equality holds for nested pairs. The nonnested estimates above use Γ , the exact additive cost in (3.3); (3.4) has the opposite direction from a bound that would replace it by γ .

4. AN EXACT CLASS OF TORSION-CONTROLLED SPECTRAL FILTERS

The critical trace comparison characterizes which continuous functions of the reciprocal eigenvalues admit a uniform linear torsion bound. Let $\psi : [0, \infty) \rightarrow \mathbb{R}$ be continuous with $\psi(0) = 0$, and define the extended nonnegative sum

$$D_\psi(D, E) = \sum_{j \geq 1} |\psi(\mu_j(D)) - \psi(\mu_j(E))|.$$

Put

$$L_\psi = \sup_{\substack{u, v \geq 0 \\ u \neq v}} \frac{|\psi(u) - \psi(v)|}{|u^r - v^r|}. \quad (4.1)$$

Equivalently, L_ψ is the Lipschitz constant of $s \mapsto \psi(s^{1/r})$. A finite constant C_ψ satisfying

$$D_\psi(A, D) \leq C_\psi(T(D) - T(A)) \quad \text{for every nested finite-volume pair } A \subset D \quad (4.2)$$

exists if and only if $L_\psi < \infty$. Writing $\lambda_B = \lambda_1(B_1)$ and $t_n = T(B_1)$, its best constant satisfies

$$\frac{L_\psi}{t_n \lambda_B^r} \leq C_{\psi, \text{opt}} \leq c_n L_\psi. \quad (4.3)$$

For sufficiency, monotonicity of the ordered reciprocal eigenvalues gives

$$|\psi(\mu_j(D)) - \psi(\mu_j(A))| \leq L_\psi(\mu_j(D)^r - \mu_j(A)^r).$$

Summation and (2.6) prove the upper bound in (4.3). Furthermore $|\psi(x)| \leq L_\psi x^r$, so

$$\sum_j |\psi(\mu_j(D))| \leq L_\psi \text{Tr } R_D^r < \infty. \quad (4.4)$$

Absolute summability is thus a consequence, not an assumption about cancellation between two spectra.

For necessity, fix $0 \leq u < v$ and let $a = \sqrt{u\lambda_B}$, $b = \sqrt{v\lambda_B}$. Take concentric balls $A = B_a$, $D = B_b$, with $A = \emptyset$ when $u = 0$. Scaling gives

$$\mu_1(A) = u, \quad \mu_1(D) = v, \quad T(D) - T(A) = t_n \lambda_B^r (v^r - u^r).$$

The first summand of (4.2) therefore yields $|\psi(v) - \psi(u)| \leq C_\psi t_n \lambda_B^r (v^r - u^r)$. Taking the supremum proves necessity and the lower bound in (4.3).

4.1. Fixed outer volume and nonmonotone filters. For $V > 0$, let B_V be a ball of volume V and put $M_V = \lambda_1(B_V)^{-1}$. The corresponding exact characterization for pairs with $|D| = V$ replaces (4.1) by

$$L_{\psi, V} = \sup_{\substack{0 \leq u, v \leq M_V \\ u \neq v}} \frac{|\psi(u) - \psi(v)|}{|u^r - v^r|} < \infty. \quad (4.5)$$

The best constant lies between $L_{\psi, V}/(t_n \lambda_B^r)$ and $c_n L_{\psi, V}$. For sufficiency, Faber–Krahn gives $\mu_j(A) \leq \mu_j(D) \leq M_V$, so the preceding proof uses only the restricted interval.

For necessity, fix $0 < u < v \leq M_V$ and the concentric balls $B_a \subset B_b$ used above. If $W = V - |B_b| > 0$, choose an integer N so large that a ball of volume W/N has first reciprocal eigenvalue below u . Place N such balls disjointly outside B_b , with union P_N , and set

$$A = B_a \dot{\cup} P_N, \quad D = B_b \dot{\cup} P_N.$$

Now $|D| = V$, the largest reciprocal eigenvalues remain u, v , and the common torsion contribution of P_N cancels. The same first-coordinate test proves the required quotient bound. When

$W = 0$ no padding is needed. Continuity of ψ supplies the endpoint $u = 0$ by letting $u \downarrow 0$. This proves the exactly fixed-volume statement. Since sufficiency holds for every $|D| \leq V$ and necessity already holds at equality, the same characterization applies under the volume upper bound.

The condition controls oscillation as well as decay. For instance, $\psi(x) = x^r \sin(1/x)$ for $x > 0$, with $\psi(0) = 0$, satisfies $|\psi(x)| \leq x^r$, but

$$\frac{\psi'(x)}{rx^{r-1}} = \sin(1/x) - \frac{1}{rx} \cos(1/x)$$

is unbounded near zero. It therefore fails (4.2) even at fixed outer volume, despite absolute summability of each transformed spectrum. In contrast, $x^r \sin(\log x)$, extended by zero, has bounded weighted derivative and is an admissible nonmonotone filter. For powers $\psi(x) = x^q$, the global condition holds precisely when $q = r$; on the fixed-volume interval it holds for $q \geq r$, with $L_{\psi,V} = (q/r)M_V^{q-r}$.

For arbitrary pairs, the critical coordinate comparison gives

$$D_\psi(D, E) \leq c_n L_\psi \Gamma(D, E), \quad (4.6)$$

and the restricted constant can be used when both volumes are at most V . Although (4.4) makes each $\psi(R_D)$ trace class, these estimates concern ordered scalar coordinates and scalar traces. They do not identify the trace norm of $\psi(R_D) - \psi(R_E)$, whose operators need not commute.

4.2. Heat traces and spectral thresholds. For a differentiable filter on $(0, \infty)$, continuous at zero, the change of variables $s = x^r$ and the mean value theorem give

$$L_\psi = \sup_{x>0} \frac{|\psi'(x)|}{rx^{r-1}}, \quad (4.7)$$

whenever this supremum is finite; the equality follows also in the reverse direction by difference quotients. For $t > 0$, take $\psi_t(0) = 0$ and $\psi_t(x) = e^{-t/x}$ for $x > 0$. Then

$$\frac{\psi'_t(x)}{rx^{r-1}} = \frac{t^{-r}}{r} (t/x)^{r+1} e^{-t/x},$$

whose maximum occurs at $t/x = r + 1$. Hence

$$0 \leq Z_D(t) - Z_A(t) \leq c_n \frac{(r+1)^{r+1}}{re^{r+1}} t^{-r} \delta, \quad A \subset D, \quad (4.8)$$

and, for arbitrary pairs,

$$\sum_j |e^{-t\lambda_j(D)} - e^{-t\lambda_j(E)}| \leq c_n \frac{(r+1)^{r+1}}{re^{r+1}} t^{-r} \Gamma(D, E). \quad (4.9)$$

The coefficient in dimension two is $27/(\pi e^3)$.

The continuous filters in this class also distinguish stability of smoothed spectral quantities from stability of a sharp count. Fix $a > 0$ and let $s_0 = \sqrt{a\lambda_B}$. For $A_\varepsilon = B_{s_0-\varepsilon}$ and $D_\varepsilon = B_{s_0+\varepsilon}$, the first reciprocal eigenvalue crosses a , while the second remains below a for sufficiently small $\varepsilon > 0$. Thus the number of Laplacian eigenvalues at most $1/a$ changes from zero to one, whereas $T(D_\varepsilon) - T(A_\varepsilon) \rightarrow 0$. Even within smooth connected nested domains in one bounded ball, a sharp count therefore has no uniform torsion modulus tending to zero without separation from its spectral threshold.

4.3. Regularized Fredholm determinants and their exact order range. Let m be a positive integer and $z > 0$. For a nonempty finite-volume open set D , its inverse belongs to $\mathcal{S}_m$ exactly when $m > d$. Sufficiency follows from (2.9); necessity follows from the interior-cube argument in Section 2.5. Assume $m > d$ and define

$$\psi_{m,z}(x) = \log(1 + zx) + \sum_{k=1}^{m-1} \frac{(-1)^k (zx)^k}{k}, \quad x \geq 0. \quad (4.10)$$

The standard regularized Fredholm convention [6] is

$$\det_m(I + zR_D) = \prod_j \left[(1 + z\mu_j(D)) \exp \left(\sum_{k=1}^{m-1} \frac{(-1)^k (z\mu_j(D))^k}{k} \right) \right]. \quad (4.11)$$

To verify convergence and fix the sign, differentiation gives

$$\psi_{m,z}(x) = (-1)^{m-1} z^m \int_0^x \frac{u^{m-1}}{1+zu} du, \quad |\psi_{m,z}(x)| \leq \frac{z^m x^m}{m}. \quad (4.12)$$

Thus $\sum_j |\psi_{m,z}(\mu_j(D))| < \infty$, and the product in (4.11) converges to a strictly positive number, independently of ordering. In fact $\sum_j |e^{\psi_{m,z}(\mu_j(D))} - 1| < \infty$, since the summable logarithms form a bounded sequence. Therefore

$$(I + zR_D) \exp \left(\sum_{k=1}^{m-1} \frac{(-1)^k (zR_D)^k}{k} \right) - I$$

is trace class and its ordinary Fredholm determinant is (4.11). Its real logarithm is

$$F_{m,z}(D) = \log \det(I + zR_D) = \sum_j \psi_{m,z}(\mu_j(D)). \quad (4.13)$$

The regularizing cancellation takes place inside $\psi_{m,z}$ before summation; separate lower-power traces need not exist. Zero-extension kernel directions contribute factors one.

Convergence is weaker than linear torsion stability. Put $\alpha = m - r$. Formula (4.7) gives

$$L_{m,z} = \frac{z^r}{r} \sup_{y>0} \frac{y^\alpha}{1+y}.$$

This supremum is finite precisely for $0 \leq \alpha \leq 1$. Define

$$b_\alpha = \begin{cases} 1, & \alpha = 0 \text{ or } \alpha = 1, \\ \alpha^\alpha (1-\alpha)^{1-\alpha}, & 0 < \alpha < 1. \end{cases}$$

For $0 < \alpha < 1$ the maximizing point is $y = \alpha/(1-\alpha)$; at the two endpoints the supremum is one. It follows that for every integer $r \leq m \leq r+1$,

$$0 \leq (-1)^{m-1} (F_{m,z}(D) - F_{m,z}(A)) \leq \frac{c_n}{r} b_{m-r} z^r \delta, \quad A \subset D. \quad (4.14)$$

The absolute arbitrary-pair difference has the same upper bound with δ replaced by $\Gamma(D, E)$; so does the sum of absolute differences of the individual ordered logarithmic factors. Indeed, (4.12) makes the filter monotone with sign $(-1)^{m-1}$, so the nested trace difference has no cancellation.

These integer orders are exactly those admitting a domain-independent linear torsion estimate for the scalar logarithm. To prove necessity directly at the trace level, let $\beta = \mu_1(B_1)$. If $d < m < r$, then

$$|F_{m,z}(B_\varepsilon)| \geq |\psi_{m,z}(\beta\varepsilon^2)| \sim \frac{z^m \beta^m}{m} \varepsilon^{2m}, \quad T(B_\varepsilon) = t_n \varepsilon^{2r}.$$

Their ratio diverges as $\varepsilon \downarrow 0$. If $m > r+1$, the asymptotic formula

$$|\psi_{m,z}(x)| \sim \frac{z^{m-1}}{m-1} x^{m-1} \quad (x \rightarrow \infty)$$

and the first coordinate on B_L , $L \rightarrow \infty$, give a ratio bounded below by a positive constant times $L^{2(m-1-r)}$. All logarithmic factors have the same sign, which justifies both first-coordinate lower bounds. Empty inner domains already suffice for these tests.

Under $|D| \leq V$, the reciprocal range is restricted to $[0, M_V]$. The derivative ratio is then bounded exactly when $m \geq r$, and one may use

$$L_{m,z;V} \leq \frac{z^m}{r} M_V^{m-r}. \quad (4.15)$$

The excluded orders $d < m < r$ fail even when $|D| = V$ exactly and the inner domain is nonempty. Choose disjoint balls C_ε and B_ε with $|C_\varepsilon| = V - |B_\varepsilon|$ and set $A_\varepsilon = C_\varepsilon$, $D_\varepsilon = C_\varepsilon \dot{\cup} B_\varepsilon$. They fit in one fixed bounded cube for small ε . Torsion and the absolutely convergent sum (4.13) are additive on disjoint components, leaving exactly the diverging small-ball ratio above.

Consequently, if $n = 2k$, the smallest convergent order is $k + 1$, the globally stable orders are $k + 1, k + 2$, and all orders $m \geq k + 1$ are stable under a volume bound. If $n = 2k + 1$, these three answers are respectively $k + 1$, $k + 2$, and $m \geq k + 2$. In particular, the smallest stable choice is $m = \lceil r \rceil$, which need not be the smallest convergent choice. In the plane, (4.14) with $m = r = 2$ reads

$$0 \leq \log \det_2(I + zR_A) - \log \det_2(I + zR_D) \leq \frac{z^2}{\pi} \delta. \quad (4.16)$$

At $z = 0$ all products equal one and all logarithmic differences vanish. These are regularized Fredholm determinants of bounded resolvent perturbations; their construction requires neither a spectral-zeta continuation nor a zeta-regularized Laplacian determinant on the domain.

5. LOW-ENERGY COMPARISON AND THE TWO FORMS OF SPECTRAL SURGERY

We now turn to finite geometric replacements in the plane. The whole-spectrum estimates and the construction below use the same torsion defect, but they serve different purposes. For a prescribed energy window, the compression of a resolvent difference has norm and trace bounded *linearly* in the torsion loss. This linear estimate allows removed area to pay for spectral error, or for a loss of rank. Using only a whole-operator Schatten bound would lose this quantitative balance. We give the low-energy argument explicitly before applying it to the finite catalogues.

5.1. A linear comparison on the entire low-energy subspace. Let $A \subset D$ be bounded open subsets of $\mathbb{R}^2$, with $D \neq \emptyset$. We allow $A = \emptyset$ temporarily. The Dirichlet realization is always the one associated with H_0^1 of the actual open set. In particular, a slit is not removed merely because it has zero area. Extend the resolvent of A by zero to $L^2(D)$, and put

$$R_D = (-\Delta_D)^{-1}, \quad R_A = (-\Delta_A)^{-1}, \quad S = R_D - R_A,$$

with $R_\emptyset = 0$. These are compact self-adjoint operators; extension may introduce an infinite-dimensional zero eigenspace. Write

$$\begin{aligned} w_D &= R_D 1, & T(D) &= \int_D w_D \, dx, \\ \mathcal{E}(D) &= -\frac{1}{2} T(D), & \Delta &= \mathcal{E}(A) - \mathcal{E}(D) = \frac{T(D) - T(A)}{2}. \end{aligned} \quad (5.1)$$

Here and below functions on a smaller open set are extended by zero when they occur in an integral on a larger one.

Two positivity properties of S will be used separately. First,

$$\langle f, R_D f \rangle = \sup_{v \in H_0^1(D)} \left(2\langle f, v \rangle - \int_D |\nabla v|^2 \, dx \right)$$

and the inclusion $H_0^1(A) \subset H_0^1(D)$ show that S is positive semidefinite. Second, weak comparison gives

$$f \geq 0 \implies Sf \geq 0 \text{ almost everywhere.} \quad (5.2)$$

For completeness, both resolvent solutions for such f are nonnegative. The test function $(R_A f - R_D f)_+$ belongs to $H_0^1(A)$: approximate $R_A f$ by compactly supported functions in A , take the positive part after subtracting the nonnegative zero extension of $R_D f$, and use continuity of truncation in H^1 . Subtracting the two weak equations and testing with this function proves that its gradient vanishes. This proves (5.2) even for rough or disconnected sets. Positive semidefiniteness alone would not imply this order property.

If v is a bounded real function, order preservation gives

$$|Sv| \leq S|v| \leq \|v\|_\infty S1.$$

Since $S1 = w_D - w_A \geq 0$, it follows that

$$0 \leq \langle v, Sv \rangle \leq \|v\|_\infty^2 \int_D (w_D - w_A) \, dx = 2\|v\|_\infty^2 \Delta. \quad (5.3)$$

All the estimates can therefore be obtained without selecting a pointwise representative of a difference of Green kernels.

Choose real orthonormal Dirichlet eigenfunctions u_i on D , ordered with multiplicity, and let $V_j = \text{span}\{u_1, \dots, u_j\}$. At a multiple endpoint any required number of orthonormal states may be chosen. The Dirichlet heat semigroup is dominated by the free heat semigroup on every open set [15]. Its diagonal expansion consequently gives, for almost every $x \in D$ and $t > 0$,

$$\rho_j(x) := \sum_{i=1}^j u_i(x)^2 \leq e^{t\lambda_j(D)} p_D(t; x, x) \leq \frac{e^{t\lambda_j(D)}}{4\pi t}.$$

Taking $t = \lambda_j(D)^{-1}$, and applying Cauchy–Schwarz to the finite expansion of a function in V_j , yields

$$\rho_j \leq \frac{e\lambda_j(D)}{4\pi}, \quad \|v\|_\infty^2 \leq \frac{e\lambda_j(D)}{4\pi} \|v\|_2^2 \quad (v \in V_j). \quad (5.4)$$

This estimates the whole spectral projector, rather than adding separate bounds for the suprema of its eigenfunctions.

Let P_j denote the orthogonal projection onto V_j . Equations (5.3) and (5.4), together with positivity, give

$$\|P_j S P_j\|_{L^2 \rightarrow L^2} \leq \frac{e\lambda_j(D)}{2\pi} \Delta. \quad (5.5)$$

On V_j , the quadratic form of R_D is bounded below by $\lambda_j(D)^{-1}$ times the squared norm. Descending compact-operator min–max therefore shows that

$$\frac{1}{\lambda_j(A)} \geq \frac{1}{\lambda_j(D)} - \frac{e\lambda_j(D)}{2\pi} \Delta \quad (5.6)$$

whenever the expression on the right is positive. More precisely, min–max first bounds the j -th descending eigenvalue of the extended operator R_A by that expression. Positivity then proves the existence of j positive eigenvalues and justifies the reciprocal notation. Thus the argument also excludes an empty A when a positive bound has been established; nonemptiness and rank have not been assumed in advance.

The compressed-resolvent argument is already present in [8, Theorem 3.4 and Lemma 3.5]. Its use in spectral surgery, and simultaneous improvement of an initial spectrum, are established antecedents [9, 11]. The order comparison and the classical heat-projector bound remove the extra index factors that arise when eigenfunction suprema are estimated separately. We use their combination as an input to the quantitative construction. Its energy-dependent coefficient serves a different purpose from a dimension-only whole-resolvent estimate.

5.2. A trace estimate with the same energy dependence. There is a useful companion to (5.5). For every bounded real f , positivity preservation implies

$$|Sf|^2 \leq S1 S(f^2) \quad \text{almost everywhere.} \quad (5.7)$$

Indeed, apply S to the nonnegative function $(f - z)^2$ for each rational z . Outside one common null set, $S(f^2) - 2zSf + z^2S1 \geq 0$ for all rational, hence all real, z . Its discriminant gives (5.7); if $S1 = 0$, the same polynomial forces $Sf = 0$. No division by $S1$ is needed.

Suppose $\lambda_k(D) \leq E$, and put $M = eE/(4\pi)$. Then $\rho_k \leq M$, and summing (5.7) gives

$$\sum_{i=1}^k |Su_i|^2 \leq S1 S(\rho_k) \leq M(S1)^2.$$

Pointwise Cauchy–Schwarz now yields

$$\sum_{i=1}^k u_i S u_i \leq \rho_k^{1/2} \left(\sum_{i=1}^k |S u_i|^2 \right)^{1/2} \leq M S 1.$$

Integrating, we obtain

$$0 \leq \text{Tr}(P_k S P_k) \leq \frac{eE}{2\pi} \Delta. \quad (5.8)$$

The trace is nonnegative by positive semidefiniteness, although its pointwise summands need not be nonnegative. The important feature is that summing the states has not introduced a further factor k .

6. TORSION MINIMIZATION WITH AN OPEN POSITIVITY SET

We next give the existence and perimeter arguments needed for the particular penalties used below. Let D be a bounded nonempty planar open set with compact smooth boundary, not necessarily connected, and let $c > 0$. Smooth boundary here has its usual domain meaning: locally the domain lies on one side of a smooth embedded curve. Minimize, over nonnegative $u \in H_0^1(D)$,

$$J_c(u) = \frac{1}{2} \int_D |\nabla u|^2 \, dx - \int_D u \, dx + c |\{u > 0\}|. \quad (6.1)$$

This standard torsion-plus-volume construction underlies [9, 11]. The details below ensure that the minimizer used here is an open set for every $c > 0$, rather than an unspecified quasi-open representative.

Poincare and Cauchy–Schwarz make J_c coercive. A minimizing sequence has a subsequence converging weakly in $H_0^1(D)$, strongly in $L^2(D)$, and almost everywhere. Its limit u is nonnegative. The Dirichlet integral is weakly lower semicontinuous, the linear term converges, and

$$1_{\{u>0\}} \leq \liminf_n 1_{\{u_n>0\}} \quad \text{almost everywhere.}$$

Fatou’s lemma proves lower semicontinuity of the volume term. Thus u minimizes (6.1).

To obtain continuity, fix a ball $B_r(x) \Subset D$, and let h be the harmonic function with the same Sobolev trace as u on that ball. The weak maximum principle gives $h \geq 0$. Replace u by h in the ball. Harmonic orthogonality and minimality, with $X = \|\nabla(u - h)\|_{L^2(B_r)}$, imply

$$\frac{1}{2} X^2 \leq \int_{B_r} |u - h| \, dx + c |B_r|.$$

Poincare on the ball and Cauchy–Schwarz bound the integral by $C r^2 X$. Young’s inequality therefore gives

$$X^2 \leq C(c + r^2) r^2. \quad (6.2)$$

Here and in this continuity argument C is an absolute constant, whose value does not enter any subsequent catalogue bound. For $F_x(r) = \int_{B_r(x)} |\nabla u|^2$, harmonic energy minimality and the mean-value inequality for $|\nabla h|^2$ then give

$$F_x(\rho) \leq 2(\rho/r)^2 F_x(r) + C(c + r^2) r^2, \quad 0 < \rho < r.$$

Fix $0 < \alpha < 1$, and choose $0 < \theta < 1$ with $2\theta^2 \leq \frac{1}{2}\theta^{2\alpha}$. Iteration on radii $r, \theta r, \theta^2 r, \dots$ gives $F_x(\rho) \leq C_{K,\alpha} \rho^{2\alpha}$, uniformly for centers in a compact subset $K \Subset D$ and small radii. The geometric series converges because $2 - 2\alpha > 0$. Poincare on balls gives the corresponding mean-oscillation bound

$$\int_{B_\rho(x)} |u - u_{B_\rho(x)}|^2 \, dx \leq C_{K,\alpha} \rho^{2+2\alpha}.$$

The Morrey–Campanato conclusion, obtained by telescoping averages over nested balls, supplies a locally $C^{0,\alpha}$ representative. Only this continuity, not free-boundary Lipschitz regularity, is required.

The torsion function w_D is positive in each component and, because D has smooth boundary, is continuous on $\bar{D}$ and vanishes there on ∂D . For $F(v) = \frac{1}{2} \int |\nabla v|^2 - \int v$, its weak equation gives the exact identity

$$F(v) = F(w_D) + \frac{1}{2} \|\nabla(v - w_D)\|_2^2, \quad v \in H_0^1(D).$$

Compare u with $v = \min(u, w_D)$. Their positivity sets agree up to null sets, and hence

$$J_c(u) - J_c(v) = \frac{1}{2} \|\nabla(u - w_D)_+\|_2^2.$$

Minimality forces $u \leq w_D$. Local continuity makes this a pointwise inequality in D , so the representative of u extended by zero is continuous across ∂D . Consequently

$$A = \{u > 0\}$$

is a bounded open subset of D .

For $s > 0$, the support of $(u - s)_+$ lies in the compact set $\{u \geq s\} \Subset A$. Local mollification therefore puts $(u - s)_+$ in $H_0^1(A)$. These truncations converge to u in H^1 as $s \downarrow 0$; the gradient of u is zero almost everywhere on $\{u = 0\}$. Thus $u \in H_0^1(A)$. On the support of any $\varphi \in C_c^\infty(A)$, u has a strictly positive minimum. Both signs of a sufficiently small variation $u + t\varphi$ leave the positivity set unchanged. Differentiating (6.1) gives

$$\int_A \nabla u \cdot \nabla \varphi \, dx = \int_A \varphi \, dx.$$

Density and uniqueness identify $u = w_A$. If $u = 0$, take $A = \emptyset$. For every open $B \subset D$, the zero-extended torsion function w_B is an admissible competitor and is positive in B , component by component. It follows that

$$\mathcal{E}(A) + c|A| \leq \mathcal{E}(B) + c|B| \quad \text{for every open } B \subset D. \quad (6.3)$$

We have also retained the stronger function-competitor minimality used in the next calculation.

6.1. Perimeter control by truncation. Put $a = |A|$ and compare $u = w_A$ with $(u - s)_+$ in (6.1). The exact energy difference gives

$$\frac{1}{2} \int_{\{0 < u \leq s\}} |\nabla u|^2 \, dx + c|\{0 < u \leq s\}| \leq \int_A \min(u, s) \, dx \leq sa.$$

Bounding the two nonnegative terms separately and applying Cauchy–Schwarz yields

$$\int_{\{0 < u \leq s\}} |\nabla u| \, dx \leq \sqrt{2/c} \, sa.$$

The coarea formula supplies levels $s_n \downarrow 0$ with $\text{Per}(\{u > s_n\}) \leq \sqrt{2/c} \, a$. Their indicators converge in L^1 to 1_A . Lower semicontinuity of De Giorgi perimeter proves

$$\text{Per}(A) \leq \sqrt{\frac{2}{c}} |A|. \quad (6.4)$$

This is the established truncation argument of [9, Theorem 2.2]; coarea and lower semicontinuity of perimeter are standard [16]. The perimeter in (6.4) is not being identified with the length of an arbitrary topological boundary.

6.2. Preserving a prefix, or paying for a loss of rank. First suppose that D has area one and $\lambda_m(D) \leq H$, and choose $c = H^{-2}$. With $a = |A|$ and $t = 1 - a$, domain monotonicity and comparison with D in (6.3) give

$$0 \leq \Delta \leq ct.$$

For $j \leq m$, (5.6) has the positive right-hand side

$$\frac{1 - \kappa_j t}{\lambda_j(D)}, \quad \kappa_j = \frac{e\lambda_j(D)^2}{2\pi H^2} \leq \frac{e}{2\pi} < \frac{1}{2}.$$

In particular $A \neq \emptyset$. Since $0 < a \leq 1$ and $\kappa_j \leq 1$,

$$a\lambda_j(A) \leq \frac{1-t}{1-\kappa_j t} \lambda_j(D) \leq \lambda_j(D).$$

For $S_0 = a^{-1/2}A$, planar scaling and (6.4) give the simultaneous conclusion

$$|S_0| = 1, \quad \lambda_j(S_0) \leq \lambda_j(D) \quad (1 \leq j \leq m), \quad \text{Per}(S_0) \leq \sqrt{2}H. \quad (6.5)$$

No gap between consecutive eigenvalues is involved.

The second output permits a loss of rank and will be used only for the scalar ratio $|D|\lambda_k(D)/k$. Let D again be smooth with $|D| = 1$, assume $\lambda_k(D) \leq E$, fix $0 < \eta \leq 1$, and choose

$$c = \frac{k\eta}{4E^2}.$$

For its minimizing set, let $a = |A|$, $t = 1 - a$. The trace bound (5.8) gives

$$\text{Tr}(P_k S P_k) \leq \frac{eE}{2\pi} \Delta \leq Ect.$$

Inside V_k , discard the eigenvectors of the positive compression $P_k S P_k$ whose eigenvalues are strictly greater than $s = \eta/((1+\eta)E)$. The retained dimension r is an integer, and counting the discarded eigenvalues by their contribution to the trace gives

$$r \geq k - \frac{\text{Tr}(P_k S P_k)}{s} \geq k - \frac{(1+\eta)E^2 ct}{\eta} = k \left(1 - \frac{1+\eta}{4}t\right) \geq \frac{k(1+a)}{2}.$$

On this retained subspace the quadratic form of R_A is at least $1/E - s = 1/((1+\eta)E)$. Resolvent min-max proves all assertions in

$$\begin{aligned} 0 < a \leq 1, \quad r \in \{1, \dots, k\}, \quad r \geq \frac{k(1+a)}{2}, \quad \lambda_r(A) \leq (1+\eta)E, \\ \frac{a}{r} \leq \frac{1}{k}, \quad \frac{a\lambda_r(A)}{r} \leq (1+\eta)\frac{E}{k}. \end{aligned} \quad (6.6)$$

Indeed $r \geq k/2 > 0$ proves nonemptiness, and $a/r \leq 2a/(k(1+a)) \leq 1/k$. Repeated eigenvalues and equality at the truncation threshold cause no difficulty. The same set satisfies (6.4) with the chosen c . It is the saved physical area, not a substitute spectral mass, that compensates for the discarded rank.

7. INNER REGULARIZATION AND HOLE-UNIFORM GRID COVERING

7.1. One family of compact trials for every prefix. Let Ω be any bounded nonempty open set and fix m and $b > 0$. Approximate its first m real eigenfunctions in $H_0^1(\Omega)$ by $f_1, \dots, f_m \in C_c^\infty(\Omega)$. The mass and energy Gram matrices of this family approach I and $\text{diag}(\lambda_1, \dots, \lambda_m)$, respectively. To make the simultaneous assertion explicit, choose the approximations so that both operator-norm errors are at most δ , where

$$0 < \delta \leq \frac{b}{2(1 + \lambda_m(\Omega) + b)}.$$

Every principal j -by- j compression has the same error bound. Its mass matrix is at least $(1-\delta)I$, and its energy matrix is at most $(\lambda_j(\Omega) + \delta)I$. Consequently every first- j trial span is j -dimensional and has maximal Rayleigh quotient at most

$$\frac{\lambda_j(\Omega) + \delta}{1 - \delta} \leq \lambda_j(\Omega) + b, \quad 1 \leq j \leq m.$$

Let $K \Subset \Omega$ be the union of these compact supports. Choose $\chi \in C_c^\infty(\Omega)$, equal to one on a neighborhood of K . A regular level in $(0, 1)$ gives a bounded smooth open set $\Omega' \Subset \Omega$ containing K . Thus

$$|\Omega'| \leq |\Omega|, \quad \lambda_j(\Omega') \leq \lambda_j(\Omega) + b \quad (1 \leq j \leq m). \quad (7.1)$$

This approximation is made inside the original form domain. It is therefore valid in the presence of arbitrary boundary roughness, disconnectedness, and positive-capacity slits.

7.2. Positive torsion levels preserve the perimeter budget. The minimizers above admit a more precise regularization than an arbitrary finite-perimeter set. Let $u = w_A$ be the continuous minimizer of (6.1), assume $a = |A| > 0$, and put $C_c = \sqrt{2/c}a$. Interior elliptic regularity gives $u \in C^\infty(A)$, since $-\Delta u = 1$ there. For every $t > 0$, continuity of the zero extension shows that $\{u \geq t\}$ is a compact subset of A . A positive regular superlevel $G_t = \{u > t\}$ consequently has compact smooth boundary.

The truncation estimate already proved gives, for every $s > 0$,

$$\int_0^s \text{Per}(G_t) \, dt = \int_{\{0 < u < s\}} |\nabla u| \, dx \leq C_c s.$$

Almost every positive value is regular by Sard's theorem. Hence for each $s > 0$ one can choose a positive regular $t < s$ with $\text{Per}(G_t) \leq C_c$. Choosing such values tending to zero gives smooth inner sets with area tending to a , all with the same perimeter bound. A compact smooth boundary has only finitely many components, so these are precisely the smooth sets needed for the planar covering argument below.

The convergence is also spectral. Given a finite prefix and a positive allowance, choose one simultaneous compact trial family in A as above. Its combined support K satisfies $\min_K u > 0$. Every sufficiently small positive superlevel contains all those trials. Inclusion in A and min-max therefore give

$$\lambda_j(A) \leq \lambda_j(G_t) \leq \lambda_j(A) + r \quad (1 \leq j \leq m)$$

along sufficiently small selected levels. In particular, every fixed eigenvalue converges to its value on A . Nestedness of the selected sequence is unnecessary.

After any dilation by $\ell > 0$, this proves the following form used throughout the construction: for every m and $r > 0$ there is a bounded smooth open set V such that

$$\begin{aligned} \bar{V} &\subset \ell A, & |V| &\leq \ell^2 a, & \text{Per}(V) &\leq \ell \sqrt{2/c} a, \\ \lambda_j(\ell A) &\leq \lambda_j(V) \leq \lambda_j(\ell A) + r & (1 \leq j \leq m). \end{aligned} \tag{7.2}$$

The area can additionally be made arbitrarily close to $\ell^2 a$ from below. For the normalized prefix output $S_0 = a^{-1/2}A$, the perimeter bound is at most $\sqrt{2}H$. For the area-rank output no dilation is made, and it is $\sqrt{2/c}a$.

This regularization is internal to the actual positivity set; it does not fill a slit or charge an extra area allowance. Its level may depend on the source and the trial supports without a known rate. The catalogue requires only the uniform area and perimeter outputs, not an algorithm reconstructing the regularized source.

7.3. A lower bound and the active components. We shall use the elementary planar consequence of the Li–Yau inequality [23],

$$|\Omega| \lambda_j(\Omega) \geq 2\pi j, \quad j \geq 1. \tag{7.3}$$

Here is its Fourier proof on the precise open-set class in use. Extend an orthonormal first- j eigenfamily by zero and take $\hat{u}(\xi) = (2\pi)^{-1} \int e^{-ix \cdot \xi} u(x) \, dx$. Bessel's inequality and Plancherel give, for $f = \sum_{i \leq j} |\hat{u}_i|^2$,

$$0 \leq f \leq M := \frac{|\Omega|}{4\pi^2}, \quad \int_{\mathbb{R}^2} f \, d\xi = j, \quad \int_{\mathbb{R}^2} |\xi|^2 f \, d\xi = \sum_{i \leq j} \lambda_i(\Omega).$$

Choose R with $M\pi R^2 = j$. The product $(|\xi|^2 - R^2)(f - M1_{\{|\xi| < R\}})$ is nonnegative. After integration and cancellation of the equal masses this gives

$$\sum_{i \leq j} \lambda_i(\Omega) \geq M \frac{\pi R^4}{2} = \frac{2\pi j^2}{|\Omega|}.$$

Bounding the sum above by $j\lambda_j(\Omega)$ proves (7.3). No smoothness or connectedness has been used.

Suppose a smooth bounded open set V has $\lambda_m(V) \leq F$. Delete every whole component whose first eigenvalue is strictly greater than F , and call the resulting union G . The Dirichlet operator is a direct sum over the open components. Removed components contribute no state at or below F , so all the first m ordered eigenvalues, including endpoint multiplicities, are unchanged. If q is the number of retained components, (7.3) gives

$$\lambda_j(G) = \lambda_j(V) \quad (1 \leq j \leq m), \quad q \leq \frac{|G|F}{2\pi} \leq \frac{|V|F}{2\pi}. \quad (7.4)$$

Area and perimeter do not increase. Compact smooth boundary ensures that only finitely many components and boundary curves occur.

7.4. Covering without a charge for the number of holes. Let G have compact smooth boundary, q open components, and perimeter p . For $t > 0$ put $G_t = \{x : \text{dist}(x, G) < t\}$. We first establish

$$|G_t| - |G| \leq (2 + \pi)tp + \pi qt^2. \quad (7.5)$$

The number of holes does not appear in this estimate.

A compact rectifiable curve of length ℓ has a radius- t neighborhood of area at most $2t\ell + \pi t^2$. For a polygonal path, begin with the radius- t disk at its initial point. Adding a segment of length a adds at most $2ta$, because the disk at the starting endpoint is already covered. Induction proves the assertion for a polygonal path, including a closed one. An inscribed polygonal approximation with subarcs of length at most δ has length at most ℓ and Hausdorff distance at most δ from the curve. Apply the polygonal bound at radius $t + \delta$ and let $\delta \downarrow 0$.

Each component of G has one outer Jordan boundary; its remaining boundary curves bound holes. An outer curve of length ℓ costs at most $2t\ell + \pi t^2$, and there are exactly q such curves. For a hole with boundary length $\ell > t$, its part within distance t of that boundary costs at most

$$2t\ell + \pi t^2 \leq (2 + \pi)t\ell.$$

If instead $\ell \leq t$, pay for the whole Jordan interior. Each coordinate span of the curve is at most $\ell/2$, since both arcs joining a coordinate minimum to a maximum have length at least that span. The hole lies in the resulting bounding box, and therefore has area at most

$$\ell^2/4 \leq t\ell/4 \leq (2 + \pi)t\ell.$$

Every point of $G_t \setminus G$ lies in one of these charged regions: choose a component at distance less than t , and follow a segment to a nearby point in that component until it crosses the relevant boundary curve. Other components inside a hole merely cause overcounting, which is harmless. Summing the charges proves (7.5). No reach, curvature bound, or separation between boundary curves has been used.

Fix a square grid of pitch $h > 0$. Select every closed grid square that meets $\overline{G}$, including all squares incident to a contact on a grid edge or vertex, and let U be the interior of their union. Every point of $\overline{G}$ is an interior point of that union; compactness even gives a positive distance to U^c . Every selected cell is within distance $\sqrt{2}h$ of $\overline{G}$, so $U \subset G_{2\sqrt{2}h}$, with the factor two ensuring strict neighborhood containment. If N cells were selected, their interiors are disjoint and all grid boundaries have zero area. Consequently (7.5) gives

$$\overline{G} \subset U, \quad |U| = Nh^2, \quad (7.6)$$

$$|U| \leq |G| + 2\sqrt{2}(2 + \pi)h \text{Per}(G) + 8\pi qh^2, \quad \lambda_j(U) \leq \lambda_j(G) \quad (j \geq 1).$$

The last assertion is Dirichlet form inclusion. The new domain may fill holes or join nearby components, but the entire added physical area has been charged. This construction is applied only to the smooth set obtained from (7.2); no outer-tube estimate for a rough finite-perimeter representative is being assumed.

7.5. Retaining and packing physical components. The domain U is a genuine grid mask: it is the interior of a finite union of closed squares, with every common cell edge opened. Its open components are exactly the edge-connected cell components. If two closures meet only at a vertex, their open sets remain disconnected. Choose an ordered first- m orthonormal eigenfamily of U from its componentwise direct-sum eigenbasis, and retain precisely the components carrying these m states. Their union U_0 has at most m components and

$$\lambda_j(U_0) = \lambda_j(U) \quad (1 \leq j \leq m). \quad (7.7)$$

Indeed domain inclusion gives the inequality in one direction, and the retained first- j trial spans give the reverse inequality. At a multiple endpoint choose any allocation of the required component-supported states. Whole components are retained, with their whole spectra, not just a selected eigenvalue list.

Dilate by h^{-1} to obtain unit cells. An edge-connected n -cell component has bounding-box width and height at most n . If the retained components have a total of N_0 cells, translate their boxes by integer vectors and place them horizontally with one blank column between successive boxes. Their union P fits in a $2N_0$ -by- N_0 grid, since the total width is at most $N_0 + q_0 - 1 \leq 2N_0 - 1$. Componentwise translations preserve the entire Dirichlet direct-sum spectrum. Finally, for the area-one normalization $P^\# = N_0^{-1/2}P$,

$$\lambda_j(P^\#) = N_0 \lambda_j(P) = |U_0| \lambda_j(U_0). \quad (7.8)$$

The normalization is of the whole packed union, not of each component separately.

8. A SIMULTANEOUS FINITE GRID REPLACEMENT

Simultaneous improving replacements and bounded representatives are established features of spectral surgery [25, 11]. Here we make the replacement finite and retain the geometric costs explicitly. The preceding constructions give the following uniform statement. For every bounded open $\Omega \subset \mathbb{R}^2$ with $|\Omega| = 1$ and $\lambda_m(\Omega) \leq E$, $E \geq 1$, a single normalized unit-cell mask $P^\#$, with at most m components, can satisfy either of the following independently prescribed requirements:

$$\begin{aligned} \lambda_j(P^\#) &\leq \lambda_j(\Omega) + \tau \quad (1 \leq j \leq m, \ 0 < \tau \leq 1), \\ N &\leq \left\lceil 211000 E^4 \tau^{-2} \right\rceil; \end{aligned} \quad (8.1)$$

$$\begin{aligned} \lambda_j(P^\#) &\leq (1 + \varepsilon) \lambda_j(\Omega) \quad (1 \leq j \leq m, \ 0 < \varepsilon \leq 1), \\ N &\leq \left\lceil 312500 E^2 \varepsilon^{-2} \right\rceil. \end{aligned} \quad (8.2)$$

The same $P^\#$ works for every index in the selected prefix. There is no lower comparison to the source spectral vector.

Here are the constants and their derivation. Choose $r_1, r_2 \in (0, 1/4]$ and $0 < d \leq 1$, and put $H = E + r_1$. Use (7.1) to obtain a smooth $\Omega' \Subset \Omega$, of area $A' \leq 1$, with $\lambda_j(\Omega') \leq \lambda_j(\Omega) + r_1$ for $j \leq m$. The smooth unit-area set $D = (A')^{-1/2} \Omega'$ then satisfies

$$\lambda_j(D) = A' \lambda_j(\Omega') \leq \lambda_j(\Omega) + r_1, \quad \lambda_m(D) \leq H.$$

Apply (6.5), and then the positive-level regularization (7.2) with spectral allowance r_2 . It gives a smooth V with $|V| \leq 1$ and $\text{Per}(V) \leq \sqrt{2}H$. For uniform conservative constants in both catalogue constructions we shall use the weaker bounds

$$|V| \leq 1 + d/4, \quad \text{Per}(V) \leq 2\sqrt{2}H < 3H, \quad \lambda_j(V) \leq \lambda_j(\Omega) + r_1 + r_2.$$

Let $F = H + r_2 \leq 5H/4$. Retain the active components at this threshold. By (7.4), their number is at most $q \leq (1 + d/4)F/(2\pi)$, while the first m levels, area bound, and perimeter bound remain valid.

Use (7.6) with

$$h = \frac{d}{100H}.$$

Since $2\sqrt{2}(2 + \pi) < 15$, the linear area increment is at most $45Hh = 9d/20 < d/2$. The quadratic increment is at most

$$8\pi qh^2 \leq 4(1 + d/4)Fh^2 \leq \frac{25}{4}Hh^2 = \frac{d^2}{1600H} \leq d/4.$$

Using $|G| \leq 1 + d/4$, we obtain

$$|U| \leq 1 + d, \quad N \leq 10000(1 + d)H^2d^{-2}, \quad \lambda_j(U) \leq \lambda_j(\Omega) + r_1 + r_2.$$

Retain at most m physical components and pack them as above. Equations (7.7) and (7.8) prove the adjustable estimate

$$\begin{aligned} N &\leq 10000(1 + d)H^2d^{-2}, \\ \lambda_j(P^\#) &\leq (1 + d)(\lambda_j(\Omega) + r_1 + r_2), \quad 1 \leq j \leq m. \end{aligned} \tag{8.3}$$

For the additive assertion set

$$r_1 = r_2 = \tau/4, \quad d = \frac{\tau}{2E + \tau}.$$

Then $d(E + \tau/2) = \tau/2$, so the second line of (8.3) is at most $\lambda_j(\Omega) + \tau$. Moreover $H \leq 5E/4$, $1 + d \leq 3/2$, and $2E + \tau \leq 3E$. Thus the cell count is at most

$$10000 \cdot \frac{3}{2} \cdot \frac{25}{16} \cdot 9E^4\tau^{-2} = 210937.5E^4\tau^{-2} \leq 211000E^4\tau^{-2},$$

as claimed in (8.1).

For the relative assertion take

$$r_1 = r_2 = \varepsilon/4, \quad d = \varepsilon/4.$$

By (7.3), $\lambda_j(\Omega) \geq 2\pi > 1$, so

$$\frac{\lambda_j(P^\#)}{\lambda_j(\Omega)} \leq (1 + \varepsilon/4)(1 + \varepsilon/2) \leq 1 + \frac{7}{8}\varepsilon \leq 1 + \varepsilon.$$

The cell count is at most

$$10000 \cdot \frac{5}{4} \cdot \frac{25}{16} \cdot 16E^2\varepsilon^{-2} = 312500E^2\varepsilon^{-2},$$

which proves (8.2). All analytic statements allow real parameter values. Effective finite libraries take rational tolerances and rational upper energy bounds; a ceiling operation on an arbitrary noncomputable real is not part of the construction.

8.1. The exact-index optimum. Let

$$\mu_k = \inf_{\substack{\Omega \subset \mathbb{R}^2 \\ \text{bounded open} \\ |\Omega|=1}} \lambda_k(\Omega).$$

The union of k separated equal squares, of total area one, has its first k eigenvalues equal to $2\pi^2k$. Hence $\mu_k \leq 2\pi^2k < 20k$, and an arbitrarily accurate minimizing sequence may be chosen with $\lambda_k \leq 20k$. Apply (8.2) with $m = k$, $E = 20k$. The finite library of packed unit-cell masks with at most k components and at most

$$B_k(\varepsilon) = \left\lceil 125000000k^2\varepsilon^{-2} \right\rceil \tag{8.4}$$

cells therefore has normalized k -th-eigenvalue minimum

$$\mu_k \leq M_{k,\varepsilon} \leq (1 + \varepsilon)\mu_k. \tag{8.5}$$

The lower bound holds because every library member is an actual admissible domain. For the upper bound apply the common finite library to a minimizing sequence and let its remaining value error tend to zero. Neither attainment of μ_k nor stabilization of a chosen sequence of domains is required. A certified interval $[l, u]$ for $M_{k,\varepsilon}$ gives $[l/(1 + \varepsilon), u]$ for μ_k .

Retaining the full selected union is essential to this argument. Selecting only one favorable component is legitimate for an area-per-index ratio, but would in general lose the prescribed exact index and invalidate (8.5).

The quantitative content of (8.4) is its explicit dependence on both the index and the tolerance. Qualitative fixed-index computability is already derivable by combining earlier spectral surgery with finite smoothing, covering, and certified pixel spectra [11, 28]. The present construction keeps the geometric, spectral, and enumeration budgets together; it does not require an optimizer to exist.

9. A COMMON LIBRARY FOR MONOTONE SPECTRAL OBJECTIVES

The same geometric library can be chosen before the spectral objective is specified. This requires removing the external energy ceiling from the preceding construction. Monotonicity alone is not coercivity; instead we use a fixed competing spectrum.

Let $\mathcal{D}$ denote all bounded open planar sets of area one, and let $x(\Omega) = (\lambda_1(\Omega), \dots, \lambda_m(\Omega))$. The union Q_m of m separated equal squares of total area one has

$$x(Q_m) = (2\pi^2 m, \dots, 2\pi^2 m). \quad (9.1)$$

The classical Cheng–Yang consequence of Yang’s inequality gives the conservative estimate

$$\lambda_m(\Omega) \leq 3m\lambda_1(\Omega). \quad (9.2)$$

We use [12, 3] for the algebraic sequence inequality, not for any regularity of the optimizing domains.

For completeness, its underlying Yang inequality applies on the present open class. Let (u_i) be a real orthonormal eigenbasis. Multiplication by a coordinate preserves $H_0^1(\Omega)$, since Ω is bounded, and

$$-\Delta(x_\alpha u_i) = \lambda_i x_\alpha u_i - 2\partial_\alpha u_i$$

holds weakly with right side in L^2 . Hence $x_\alpha u_i$ belongs to the Dirichlet operator domain, which justifies both spectral identities below, including their absolute convergence. Put $a_{ij} = \langle x_\alpha u_i, u_j \rangle$. Spectral expansion and integration by parts, justified by H_0^1 approximation, give

$$\sum_j (\lambda_j - \lambda_i) |a_{ij}|^2 = 1, \quad \sum_j (\lambda_j - \lambda_i)^2 |a_{ij}|^2 = 4 \|\partial_\alpha u_i\|_2^2.$$

For $\Lambda = \lambda_{n+1}$, subtract the second identity multiplied by $\Lambda - \lambda_i$ from the first multiplied by $(\Lambda - \lambda_i)^2$, and sum over $i \leq n$. Terms with $i, j \leq n$ cancel in pairs. The remaining terms are nonpositive, because $\Lambda - \lambda_j \leq 0$ for $j > n$. Summing over the two coordinates yields

$$\sum_{i \leq n} (\Lambda - \lambda_i)^2 \leq 2 \sum_{i \leq n} (\Lambda - \lambda_i) \lambda_i.$$

The subsequent Cheng–Yang implication is an inequality for the positive ordered sequence. Smoothness, connectivity and simple eigenvalues are not additional hypotheses of (9.2).

If $\lambda_1(\Omega) > 2\pi^2 m$, the reference (9.1) is coordinatewise no larger than $x(\Omega)$. Otherwise (9.2) gives

$$\lambda_m(\Omega) \leq 6\pi^2 m^2 < E_m, \quad E_m := 60m^2.$$

The reference itself also lies below this ceiling. Therefore every real-valued coordinatewise nondecreasing function Φ satisfies

$$\inf_{\Omega \in \mathcal{D}} \Phi(x(\Omega)) = \inf_{\substack{\Omega \in \mathcal{D} \\ \lambda_m(\Omega) \leq E_m}} \Phi(x(\Omega)). \quad (9.3)$$

Continuity is not needed for this equality.

For rational $0 < \tau \leq 1$, use the additive replacement with

$$B(m, \tau) = \left\lceil 211000(60m^2)^4 \tau^{-2} \right\rceil. \quad (9.4)$$

Let $\mathcal{C}(m, \tau)$ be the finite set of canonical packed unions of at most m connected masks, with at most $B(m, \tau)$ cells in total, normalized as entire unions to area one. The packing and its

enumeration are described below. For every source in the right side of (9.3), some $P^\# \in \mathcal{C}(m, \tau)$ satisfies

$$x(P^\#) \leq x(\Omega) + \tau \mathbf{1} \quad (9.5)$$

coordinatewise.

Assume now that Φ is computable and continuous on the nonnegative orthant and that an effective modulus ω is specified on $[0, E_m + 1]^m$, in the sup norm. Here computability means that a supplied uniform algorithm, given rational approximation oracles for the argument coordinates and a requested positive accuracy, returns a rational interval enclosing the value with that accuracy (Type-2 computability). Pointwise computability at computable inputs, or a modulus alone, is not this assumption. Write

$$I_\Phi = \inf_{\Omega \in \mathcal{D}} \Phi(x(\Omega)), \quad M_\Phi = \min_{P^\# \in \mathcal{C}(m, \tau)} \Phi(x(P^\#)).$$

The infimum is finite, since $\Phi(0)$ is a lower bound and $\Phi(x(Q_m))$ is a finite upper bound. First use monotonicity in (9.5), and then apply the modulus between $x(\Omega) + \tau \mathbf{1}$ and $x(\Omega)$. This gives

$$I_\Phi \leq M_\Phi \leq I_\Phi + \omega(\tau). \quad (9.6)$$

It would be incorrect to use the modulus directly between $x(P^\#)$ and $x(\Omega)$: those vectors need not be close on the lower side.

Every member of the library is an admissible domain, but some members may have $\lambda_m > E_m + 1$. The assumption of global computability of Φ permits their evaluation. An explicit common box is available as well. An N -cell mask contains an open unit square, whence

$$\lambda_j(P^\#) \leq B(m, \tau) \lambda_j((0, 1)^2) < 10B(m, \tau)(1 + j^2).$$

Thus evaluation does not require deciding whether a candidate lies exactly on an energy threshold.

Suppose each candidate objective value has been enclosed in a rational interval $[l_P, u_P]$ of width at most $\eta > 0$. The continuum eigenvalue procedure in Section 13 makes this possible. Set

$$l = \min_P l_P, \quad u = \min_P u_P.$$

Then $M_\Phi \in [l, u]$ and $u - l \leq \eta$, using the candidate attaining l for the width estimate. Consequently

$$I_\Phi \in [l - \omega(\tau), u]. \quad (9.7)$$

Choose a candidate P_* attaining the least rational upper endpoint u . Comparison with a candidate attaining the true finite minimum gives $u \leq M_\Phi + \eta$, and therefore

$$\Phi(x(P_*^\#)) \leq I_\Phi + \omega(\tau) + \eta. \quad (9.8)$$

This returns an actual approximate minimizing domain as well as a value enclosure. Neither exact comparisons of spectral values nor tie-breaking between exact minimizers is required. Choose τ and η with $\omega(\tau) + \eta$ below the desired error. The general passage from effective compact approximation to computable optimization is classical; see, for example, [26]. Here its concrete input is the objective-independent spectral library with an explicit budget. The procedure terminates, but no uniform time bound for an arbitrary computable objective is asserted.

One direct application is simultaneous spectral feasibility. For rational thresholds $b_1, \dots, b_m$, put

$$\Phi_b(x) = \max_{1 \leq j \leq m} (x_j - b_j).$$

This objective is monotone and 1-Lipschitz. Under the promise

$$\inf_{\Omega \in \mathcal{D}} \Phi_b(x(\Omega)) \leq -\gamma \quad \text{or} \quad \inf_{\Omega \in \mathcal{D}} \Phi_b(x(\Omega)) \geq \gamma, \quad \gamma > 0,$$

an enclosure of total error smaller than γ distinguishes the alternatives. In the first it returns a domain satisfying $\lambda_j < b_j$ for every j ; in the second it excludes every domain satisfying all $\lambda_j \leq b_j$. No decision on the zero boundary of feasibility follows. Nonnegative weighted sums,

maxima and other computable monotone objectives are covered; arbitrary eigenvalue gaps, signed combinations and ratios are not.

10. EXACT-MASS LOCALIZATION AND A FINITE-INDEX MODULUS

For the optimal unit-area eigenvalues introduced above, set

$$\mu_j = \inf_{|\Omega|=1} \lambda_j(\Omega), \quad L = \inf_{j \geq 1} \frac{\mu_j}{j}, \quad \beta_N = \min_{1 \leq j \leq N} \frac{\mu_j}{j}. \quad (10.1)$$

Planar scaling also gives $\mu_j = \inf_{\Omega} |\Omega| \lambda_j(\Omega)$, where the latter infimum is over all bounded nonempty open planar sets. The classical Li–Yau estimate [23], in the elementary form established in (7.3), gives

$$|\Omega| \lambda_j(\Omega) \geq 2\pi j, \quad 2\pi \leq L \leq 4\pi. \quad (10.2)$$

For the last upper bound it suffices to use the unit square. Its eigenvalues are $\pi^2(p^2 + q^2)$, $p, q \in \mathbb{N}$. The number of positive lattice points in the quarter disk of radius R is bounded above by $\pi R^2/4$ and below by $\pi(R - \sqrt{2})_+^2/4$. Indeed, the southwest unit squares based at those lattice points lie in the larger quarter disk, and they cover the smaller quarter disk up to sets of measure zero. These bounds give $\lambda_j((0, 1)^2)/j \rightarrow 4\pi$.

The sequence (μ_j) is subadditive. To see the relevant scaling directly, take unit-area near-minimizers with p -th and q -th eigenvalues a and b . Scale their areas to $a/(a+b)$ and $b/(a+b)$ and place them apart. The union has area one and at least $p+q$ eigenvalues at most $a+b$. Passing to the two infima gives $\mu_{p+q} \leq \mu_p + \mu_q$. Consequently L is also the limit of μ_j/j , as in the general union framework of [13]. Subadditivity alone does not give the uniform modulus below. Our localization argument uses only the infimum in (10.1).

10.1. Sliding windows with exact mass. Fix a pitch $a > 0$ and an overlap width $0 < w \leq a$. For a translation $t \in [0, a)$, put interfaces at $t + ma$, $m \in \mathbb{Z}$. On the band of width w centered at an interface b , interpolate the cell functions on its left and right by

$$\cos\left(\frac{\pi(x - b + w/2)}{2w}\right) \quad \text{and} \quad \sin\left(\frac{\pi(x - b + w/2)}{2w}\right).$$

Between successive bands the unique cell function is one; elsewhere it is zero. Thus the function associated with the cell $(t + ma, t + (m+1)a)$ is supported in the closure of the interval $(t + ma - w/2, t + (m+1)a + w/2)$. The functions are nonnegative and Lipschitz. The convention $w = a$ simply removes the constant plateaux. In one dimension they satisfy

$$\sum_m \psi_m^2 = 1, \quad \sum_m |\psi'_m|^2 \leq \frac{\pi^2}{4w^2} \quad \text{almost everywhere.}$$

Tensoring the construction in the two coordinates, for $t \in [0, a)^2$ we obtain functions χ_p , $p \in \mathbb{Z}^2$, and open support squares $Q_p(t)$ of side $a + w$, with

$$\sum_p \chi_p^2 = 1, \quad \sum_p |\nabla \chi_p|^2 \leq \frac{\pi^2}{2w^2}. \quad (10.3)$$

Let $A = |\Omega|$, put $\Omega_p(t) = \Omega \cap Q_p(t)$, and define $S(t) = \sum_p |\Omega_p(t)|$. Only finitely many pieces are nonempty. A fixed point belongs, on average over a period, to $(a+w)/a$ support intervals in either coordinate. Fubini's theorem therefore gives the exact identity

$$\frac{1}{a^2} \int_{[0, a]^2} S(t) \, dt = A \left(1 + \frac{w}{a}\right)^2. \quad (10.4)$$

Choose a translation for which $S(t) \leq A(1 + w/a)^2$. This is a budget for the sum of the overlapping physical areas, not an assertion that the windows partition Ω .

For every $u \in H_0^1(\Omega)$ one has $\chi_p u \in H_0^1(\Omega_p(t))$. Here the boundary assertion must be made in the Sobolev sense. First approximate u in H^1 by functions in $C_c^\infty(\Omega)$. For each such function, cut its product with χ_p off in a shrinking layer along $\partial Q_p(t)$. The function χ_p vanishes at least linearly in the distance to this square boundary. The derivative of a

layer cutoff is of order the reciprocal layer width, so the additional gradient term is bounded on a layer whose measure tends to zero. All the resulting errors tend to zero in H^1 . The cut-off products have compact support in $\Omega \cap Q_p(t)$ and can be mollified there. Passing first through this approximation and then through the approximation of u proves the assertion. In particular, this operation does not remove any positive-capacity slit of the original domain.

Write $q_G(v) = \int_G |\nabla v|^2$ for the Dirichlet form. The map

$$T_t u = (\chi_p u)_p \quad \text{from } L^2(\Omega) \text{ to } \bigoplus_p L^2(\Omega_p(t))$$

is an exact isometry. For $u \in H_0^1(\Omega)$,

$$\begin{aligned} \sum_p q_{\Omega_p(t)}(\chi_p u) &= q_\Omega(u) + \int_\Omega \sum_p |\nabla \chi_p|^2 |u|^2 \, dx \\ &\leq q_\Omega(u) + \frac{\pi^2}{2w^2} \|u\|_{L^2(\Omega)}^2. \end{aligned} \quad (10.5)$$

The mixed terms cancel because $\sum_p \chi_p \nabla \chi_p = 0$. Both the physical mass and the dimension of a trial subspace are preserved before a window is selected.

For an open set G and a real threshold F , let $N_G(F) = \#\{j : \lambda_j(G) \leq F\}$. Counts in this notation include equality at the threshold. Suppose that $\lambda_k(\Omega) \leq E$, where $k \geq 1$ and $E > 0$, and put

$$F = E + \frac{\pi^2}{2w^2}.$$

The image under T_t of an ordered first- k spectral subspace is k -dimensional, and (10.5) bounds its direct-sum Rayleigh quotient by F . The min-max principle consequently gives

$$\sum_p N_{\Omega_p(t)}(F) \geq k. \quad (10.6)$$

There is no loss at a multiple eigenvalue: one takes precisely k orthonormal states in the source and uses the inclusive count in the direct sum.

Set $n_p = N_{\Omega_p(t)}(F)$ for each nonempty piece. Averaging the ratios $n_p/|\Omega_p(t)|$ with weights $|\Omega_p(t)|$ shows that some piece $U = \Omega_p(t)$ has $n = n_p \geq 1$ and

$$\frac{n}{|U|} \geq \frac{k}{S(t)} \geq \frac{k}{A(1+w/a)^2}.$$

Pieces with $n_p = 0$ remain in the area sum and cause no difficulty. Since $\lambda_n(U) \leq F$, we have obtained

$$\frac{|U| \lambda_n(U)}{n} \leq \frac{AE}{k} \left(1 + \frac{\pi^2}{2Ew^2}\right) \left(1 + \frac{w}{a}\right)^2. \quad (10.7)$$

The selected index admits an independent bound. The domain U is contained in a square Q of side $a + w$, so domain monotonicity gives $n \leq N_Q(F)$. If $(p, q) \in \mathbb{N}^2$ lies in a quarter disk of radius R , its southwest unit square $(p-1, p) \times (q-1, q)$ lies in that disk. These squares have disjoint interiors, even when the lattice point is on the circular boundary. Thus their number is at most $\pi R^2/4$. Applying this with $R = (a+w)\sqrt{F}/\pi$ proves

$$1 \leq n \leq N_Q(F) \leq \frac{(a+w)^2 F}{4\pi}. \quad (10.8)$$

The square bounds the index; it does not replace the selected domain or its spectrum.

10.2. An explicit index modulus and localized deficits. For $0 < \sigma \leq 1$, choose

$$w = \frac{\pi}{\sqrt{2\sigma E}}, \quad a = \frac{w}{\sigma}, \quad F = (1 + \sigma)E.$$

Equations (10.7)–(10.8) become

$$\frac{|U| \lambda_n(U)}{n} \leq \frac{AE}{k} (1 + \sigma)^3, \quad 1 \leq n \leq \left\lfloor \frac{\pi(1 + \sigma)^3}{8\sigma^3} \right\rfloor. \quad (10.9)$$

The side of the containing square, measured in units $F^{-1/2}$, is exactly

$$(a + w)\sqrt{F} = \frac{\pi(1 + \sigma)^{3/2}}{\sqrt{2}\sigma^{3/2}}. \quad (10.10)$$

The source index has disappeared from the new index and window-size bounds, but not from the possible geometry within the window.

For every integer $N \geq 4$, this implies

$$\begin{aligned} L \leq \beta_N \leq L \left[1 + \left(\frac{\pi}{N} \right)^{1/3} \right]^3, \\ 0 \leq \beta_N - L \leq 28\pi \left(\frac{\pi}{N} \right)^{1/3}. \end{aligned} \quad (10.11)$$

Indeed, put $\sigma = (\pi/N)^{1/3} \leq 1$. The real upper bound for the output index in (10.9) is $N(1 + \sigma)^3/8 \leq N$. For any $s > 0$, the definition of L provides a bounded open source and an index k for which $A\lambda_k(\Omega)/k < L + s$. Apply (10.9) with $E = \lambda_k(\Omega)$. The resulting integer n is at most N , and

$$\beta_N \leq \frac{\mu_n}{n} \leq \frac{|U|\lambda_n(U)}{n} < (L + s)(1 + \sigma)^3.$$

Let s tend to zero. The lower inequality follows from the definition of L . Finally, $(1 + \sigma)^3 - 1 \leq 7\sigma$ and $L \leq 4\pi$ give the absolute bound in (10.11). No infimum needs to be attained.

One can invert the real index bound without this last simplification. For $N \geq 4$, let $q = (\pi/(8N))^{1/3}$ and $\sigma = q/(1 - q)$. Then $q < 1/2$, so $0 < \sigma < 1$, and $\pi(1 + \sigma)^3/(8\sigma^3) = N$ exactly. The same argument gives the slightly sharper form

$$\beta_N \leq L \left[1 - \left(\frac{\pi}{8N} \right)^{1/3} \right]^{-3}. \quad (10.12)$$

These are estimates for the running minimum of optimal ratios, not estimates at the exact index N .

For comparison with the conjectured planar Pólya bound [27], suppose an actual source satisfies

$$\frac{|\Omega|\lambda_k(\Omega)}{k} \leq 4\pi(1 - \delta), \quad 0 < \delta < 1.$$

Choose $\sigma = \delta/8$. Since

$$\frac{(1 + \delta/8)^3 - 1}{\delta} \leq \frac{3}{8} + \frac{3}{64} + \frac{1}{512} = \frac{217}{512} < \frac{1}{2},$$

we have $(1 - \delta)(1 + \delta/8)^3 \leq (1 - \delta)(1 + \delta/2) = 1 - \delta/2 - \delta^2/2 \leq 1 - \delta/2$. Thus the output of (10.9) satisfies

$$\frac{|U|\lambda_n(U)}{n} \leq 4\pi(1 - \delta/2), \quad 1 \leq n \leq \left\lfloor 96\pi\delta^{-3} \right\rfloor. \quad (10.13)$$

Here the index estimate uses $(1 + \delta/8)^3 \leq (9/8)^3 < 3/2$. There is also a geometric normalization: since $|U|F \geq |U|\lambda_1(U) \geq 2\pi$, rescaling U to area one puts it in a square of side at most

$$\frac{a + w}{\sqrt{|U|}} \leq \sqrt{\frac{(a + w)^2 F}{2\pi}} \leq \sqrt{192\pi} \delta^{-3/2}.$$

This does not bound the number of holes or boundary features of the localized domain. A separate geometric replacement is needed for a finite catalogue.

11. A CONNECTED FINITE CATALOGUE FOR THE ALL-INDEX OPTIMUM

We next combine localization with the trace form of torsion surgery. The relevant feature of (6.6) is that the physical area removed by surgery pays for the discarded spectral dimension. This allows a smaller catalogue for the scalar all-index infimum than direct insertion of the localized energy ceiling into the full-prefix construction.

11.1. From the trace estimate to a connected grid domain. Let D be smooth and bounded, with $|D| = 1$ and $\lambda_k(D) \leq E$. Fix $0 < \eta, d \leq 1$ and set

$$c = \frac{k\eta}{4E^2}.$$

Choose the open torsion minimizer $A \subset D$ used in (6.6), and write $a = |A|$. That estimate and (6.4) give an integer r such that

$$\begin{aligned} 0 < a \leq 1, \quad \frac{k(1+a)}{2} \leq r \leq k, \\ \lambda_r(A) \leq (1+\eta)E, \quad \text{Per}(A) \leq \sqrt{\frac{2}{c}} a. \end{aligned} \tag{11.1}$$

In particular,

$$\frac{a}{r} \leq \frac{1}{k}. \tag{11.2}$$

All these assertions hold without an eigenvalue gap or a choice of a full eigenspace at the last retained level.

Apply the positive-level regularization (7.2) to A with spectral allowance $\zeta(1+\eta)E$, where $\zeta = d/4$. The resulting smooth inner domain has area at most a and perimeter at most $\sqrt{2/c}a$. We may therefore use the conservative bounds $(1+\zeta)a$ and $2\sqrt{2/c}a$, respectively. Its r -th eigenvalue is at most

$$H = (1+\zeta)(1+\eta)E.$$

Delete its components whose first eigenvalue is strictly greater than H . As in (7.4), the first r eigenvalues are unchanged, including multiplicities at H . Denote the remaining smooth domain by G and its number of components by q . The first-eigenvalue case of (10.2), applied to every component, gives

$$\begin{aligned} |G| \leq (1+\zeta)a, \quad \text{Per}(G) \leq 2\sqrt{\frac{2}{c}}a, \\ \lambda_r(G) \leq H, \quad q \leq \frac{(1+\zeta)aH}{2\pi}. \end{aligned} \tag{11.3}$$

Put $C = 36/7 > 2 + \pi$ and choose the grid pitch

$$h = \frac{d\sqrt{c}}{32C}. \tag{11.4}$$

Select every closed h -grid cell meeting $\overline{G}$, including all cells incident at an edge or a vertex, and let U be the interior of their union. The actual open containment $\overline{G} \subset U$ and the hole-uniform bound (7.6) yield

$$|U| - |G| \leq 2\sqrt{2}(2+\pi)h \text{Per}(G) + 8\pi qh^2. \tag{11.5}$$

For clarity, both increments can be paid using the original physical area a . The first is at most

$$2\sqrt{2}(2+\pi)h 2\sqrt{2/c}a = \frac{2+\pi}{4C}ad \leq \frac{ad}{4}.$$

For the second, (11.3) gives

$$\begin{aligned} 8\pi qh^2 &\leq 4(1+\zeta)aHh^2 \\ &\leq \frac{25}{2048C^2}aEcd^2 \leq \frac{ad}{2}. \end{aligned}$$

In the middle inequality we used $\zeta = d/4 \leq 1/4$ and $1+\eta \leq 2$. For the last one, $E \geq 2\pi k$ implies $Ec = k\eta/(4E) \leq \eta/(8\pi) < 1$, and $d \leq 1$. Together with the bound $|G| \leq (1+d/4)a$, this proves

$$\begin{aligned} |U| &\leq (1+d)a, \quad \lambda_r(U) \leq H, \\ N_{\text{cells}} &= \frac{|U|}{h^2} \leq \frac{8192C^2E^2}{k\eta d^2}. \end{aligned} \tag{11.6}$$

Indeed, $h^2 = d^2 c / (1024 C^2)$ and $(1+d)a \leq 2$. This computation includes the area of any holes filled or components joined by the outer grid. Using (11.2), we obtain

$$\frac{|U| \lambda_r(U)}{r} \leq (1+d)(1+d/4)(1+\eta) \frac{E}{k}. \quad (11.7)$$

For this scalar ratio it is now permissible to select a single component. Write U_i for the open components of U and allocate an ordered first- r eigenfamily among them. Let r_i be the number allocated to U_i ; at a multiple endpoint allocate only the required number of states. Then $\sum_i r_i = r$ and, whenever $r_i > 0$, $\lambda_{r_i}(U_i) \leq \lambda_r(U)$. Consequently

$$\sum_{r_i > 0} \frac{r_i}{r} \frac{|U_i| \lambda_{r_i}(U_i)}{r_i} \leq \frac{\lambda_r(U)}{r} \sum_{r_i > 0} |U_i| \leq \frac{|U| \lambda_r(U)}{r}.$$

Some component therefore has no larger ratio. It contains at most N_{cells} cells and has an index $j \leq r \leq k$. The occupied cells of an open component are edge-connected; isolated vertex contacts do not connect open components. After dilation by h^{-1} this component is a unit-grid domain P and planar scaling preserves its ratio. A connected N -cell mask uses at most N consecutive columns and at most N consecutive rows, so an integer translation places it in an N by N grid. We have proved the explicit intermediate bounds

$$\begin{aligned} \#P &\leq \frac{8192 C^2 E^2}{k \eta d^2}, \quad 1 \leq j \leq k, \\ \frac{|P| \lambda_j(P)}{j} &\leq (1+d)(1+d/4)(1+\eta) \frac{E}{k}, \end{aligned} \quad (11.8)$$

where $\#P$ denotes the number of occupied unit cells. Selecting one component here would not preserve an exact prescribed index or a whole spectral prefix; those tasks require the multi-component catalogue instead.

11.2. A smooth localized source with its full error payment. Fix $0 < \varepsilon \leq 1$. Since $L > 0$, choose a bounded nonempty open source Ω_0 and an integer K with

$$\frac{|\Omega_0| \lambda_K(\Omega_0)}{K} < L(1 + \varepsilon/64).$$

Apply (10.9) with $\sigma = \varepsilon/8$. It supplies a bounded open V and an integer k such that

$$\frac{|V| \lambda_k(V)}{k} \leq L(1 + \varepsilon/64)(1 + \varepsilon/2), \quad 1 \leq k \leq \lfloor 96\pi\varepsilon^{-3} \rfloor. \quad (11.9)$$

The inequalities used here are valid also at $\varepsilon = 1$:

$$(1 + \varepsilon/8)^3 \leq 1 + \varepsilon/2, \quad (1 + \varepsilon/8)^3 < 3/2.$$

The set V need not be smooth. Use the simultaneous compact-trial construction (7.1) to make a same-index smooth inner replacement with allowance

$$s = \frac{\pi \varepsilon k}{32|V|}.$$

It gives a smooth bounded open set V' with $\overline{V'} \subset V$ and

$$A' = |V'| \leq |V|, \quad \lambda_k(V') \leq \lambda_k(V) + \frac{\pi \varepsilon k}{32|V|}. \quad (11.10)$$

Thus the index is unchanged and the regularization cost has been paid explicitly.

Normalize $D = (A')^{-1/2} V'$ and set $E = \lambda_k(D) = A' \lambda_k(V')$. Then $|D| = 1$, and (11.9)–(11.10), together with $L \geq 2\pi$, imply

$$\begin{aligned} \frac{E}{k} &\leq L[(1 + \varepsilon/64)(1 + \varepsilon/2) + \varepsilon/64] \\ &= L \left[1 + \frac{17}{32} \varepsilon + \frac{\varepsilon^2}{128} \right] \leq L \left(1 + \frac{69}{128} \varepsilon \right) < 8\pi. \end{aligned} \quad (11.11)$$

The final inequality follows from $L \leq 4\pi$ and $4(1 + 69/128) = 197/32 < 8$. Thus both the localized index and its area-normalized energy are controlled without any geometric constant of the original source.

11.3. The finite minimum and its rational budgets. Apply (11.8) to this smooth unit-area source, taking

$$\eta = d = \varepsilon/10.$$

For $0 \leq \varepsilon \leq 1$, expansion gives

$$(1 + \varepsilon/10)^2(1 + \varepsilon/40) = 1 + \frac{9}{40}\varepsilon + \frac{3}{200}\varepsilon^2 + \frac{1}{4000}\varepsilon^3 \leq 1 + \frac{961}{4000}\varepsilon.$$

Combining this with (11.11) yields a connected unit-grid domain P and an integer $j \leq k$ for which

$$\frac{|P|\lambda_j(P)}{j} \leq L \left(1 + \frac{465317}{512000}\varepsilon\right) < (1 + \varepsilon)L. \quad (11.12)$$

In obtaining the displayed coefficient we used $\varepsilon^2 \leq \varepsilon$ after multiplying the two linear majorants. Explicitly,

$$\frac{69}{128} + \frac{961}{4000} + \frac{69 \cdot 961}{128 \cdot 4000} = \frac{465317}{512000} < 1.$$

For the cell count, (11.9) and (11.11) give

$$\frac{E^2}{k} < (8\pi)^2 k \leq 6144\pi^3 \varepsilon^{-3}.$$

Since $\eta d^2 = \varepsilon^3/1000$, the bound in (11.8) becomes

$$\#P \leq 50331648000 C^2 \pi^3 \varepsilon^{-6} < 2^{46} \varepsilon^{-6}. \quad (11.13)$$

The last comparison can be checked with rational arithmetic alone:

$$50331648000 \left(\frac{36}{7}\right)^2 \left(\frac{22}{7}\right)^3 = \frac{694567078723584000}{16807} < 70368744177664 = 2^{46}.$$

Also $96\pi < 302$. Define the integer budgets

$$B_\varepsilon = \lceil 2^{46} \varepsilon^{-6} \rceil, \quad J_\varepsilon = \lceil 302 \varepsilon^{-3} \rceil. \quad (11.14)$$

For rational positive ε these are computed exactly by integer arithmetic. The construction above gives $\#P \leq B_\varepsilon$ and $j \leq k \leq \lfloor 96\pi \varepsilon^{-3} \rfloor \leq J_\varepsilon$; no rounding of a real lower rank bound into an artificial spectral index is involved.

Let $\mathcal{C}_\varepsilon$ be the finite family of nonempty edge-connected masks of at most B_ε unit cells, translated to lie in a B_ε by B_ε grid. A mask denotes the interior of the union of its closed cells. Its Dirichlet spectrum is the continuum spectrum on that open domain, including at re-entrant corners, holes and boundary point contacts. Define

$$M_\varepsilon = \min \left\{ \frac{|P|\lambda_j(P)}{j} : P \in \mathcal{C}_\varepsilon, \quad 1 \leq j \leq J_\varepsilon \right\}. \quad (11.15)$$

The family is nonempty, since it contains a single square. Every entry in this finite minimum is at least L by (10.1). Conversely, the connected domain obtained in (11.12) belongs to the family after an integer translation. Therefore

$$L \leq M_\varepsilon \leq (1 + \varepsilon)L, \quad 0 < \varepsilon \leq 1. \quad (11.16)$$

This is a uniform finite geometric and finite-index reduction of the all-index infimum. It concerns the continuum eigenvalues of actual domains, not eigenvalues of a discrete grid Laplacian. The inverse-sixth-power bound is a cell-count estimate; neither the number of masks nor the running time of exhaustive certification is asserted to be polynomial. Computing the finite minimum still requires the fixed-domain spectral enclosures described separately.

As a direct finite-deficit consequence, if any bounded open source and index satisfy $|\Omega|\lambda_k(\Omega)/k \leq 4\pi(1 - \delta)$, with $0 < \delta < 1$, then $L \leq 4\pi(1 - \delta)$. Take $\varepsilon = \delta/2$ in (11.14)–(11.16). Since the finite minimum is attained by one of its actual entries, there are a connected unit-grid domain P and an integer j with

$$\begin{aligned} \#P &\leq \left\lceil 2^{52}\delta^{-6} \right\rceil, & 1 \leq j &\leq \left\lceil 2416\delta^{-3} \right\rceil, \\ \frac{|P|\lambda_j(P)}{j} &\leq 4\pi(1 - \delta/2). \end{aligned} \tag{11.17}$$

Indeed $(1 + \delta/2)(1 - \delta) = 1 - \delta/2 - \delta^2/2 \leq 1 - \delta/2$. This is conditional capture of a hypothetical violation, not an assertion that a violation exists, and it does not decide the equality $L = 4\pi$.

12. ENCODING THE GEOMETRIC LIBRARIES AND THEIR BOUNDARY BUDGETS

12.1. Cell codes and canonical component packing. The masks need not be enumerated as all bitmaps in a large containing rectangle. A standard lattice-animal encoding gives a smaller sufficient library. Tree encodings and substantially sharper polyomino enumeration estimates have a classical history; see Klarner and Rivest [20]. We need only the following elementary bounds.

For an edge-connected n -cell shape, choose its lexicographically first cell as root and translate that root to $(0, 0)$. Run a deterministic depth-first traversal of the cell-adjacency graph, with a fixed compass-neighbor priority. Record every traversal of a tree edge and its return. The word has exactly $2(n - 1)$ letters over four directions. Its visited cell set determines the entire shape, including every adjacency not traversed by the tree. Thus the number a_n of translation classes satisfies

$$a_n \leq 4^{2(n-1)} = 16^{n-1}, \quad \sum_{n=1}^B a_n \leq \frac{16^B - 1}{15}.$$

The decoded domain opens all shared edges. Replacing the shape by the traversed tree and leaving other edges as Dirichlet slits would not be this encoding.

For q ordered components of positive sizes summing to n , the number of size lists is $\binom{n-1}{q-1}$. Multiplying the connected-shape bounds and summing gives

$$\sum_{q=1}^{\min(m,n)} \binom{n-1}{q-1} 16^{n-q} \leq 17^{n-1}.$$

It follows that component lists with at most B cells and at most m components number no more than

$$\frac{17^B - 1}{16}. \tag{12.1}$$

Permutations can be removed by sorting their codes, but this is unnecessary for the upper bound.

Translate each component bounding box to the origin, then place the boxes side by side with one blank grid column between successive boxes. If their widths are w_i , a component with n_i cells has $w_i \leq n_i$ and height at most n_i . The packed union therefore fits in a $2B$ by B box. The blank columns add no domain area. Translation on each component is a unitary identification of the Dirichlet direct sums, preserving the complete spectrum and all multiplicities. Only after packing is the entire union normalized to area one.

These encodings are effective: enumerate finite words, decode, and reject those failing the required size, adjacency and canonical-code tests. They bound geometric candidates, not the cost of certifying their spectra. A connected catalogue with indices $j \leq J$ has at most $J(16^B - 1)/15$ domain–index pairs. The factor J cannot be omitted from that count.

The cell budgets ensure that the search is finite. A complementary count, with a smaller inverse-tolerance power in its logarithm, is available because the outer-grid construction also controls the number of exposed cell edges. This is a geometric counting estimate, not a bound on the time needed to certify the continuum spectra of all the candidates.

12.2. Exposed edges and the holes visible to the grid. Use the construction of (7.6): G is a bounded smooth planar open set with q components and perimeter p , every closed h -cell meeting $\overline{G}$ is selected, and U is the interior of their union. Let $b(U)$ be the number of grid edges adjacent to exactly one selected cell. Each edge is counted once, in units of the grid pitch. We claim that

$$b(U) \leq 96 \frac{p}{h} + 64q. \quad (12.2)$$

Consider an exposed edge between a selected closed cell Q and an unselected closed cell Q' . The selection convention is important: $Q' \cap \overline{G} = \emptyset$, even at its vertices. Choose $x \in Q \cap \overline{G}$ and join it by a segment to the center of Q' . The union $Q \cup Q'$ is a rectangle. The last point y of the segment in $\overline{G}$ exists by compactness and lies in Q , on a boundary curve γ of G . The remaining segment lies in the component C of $\mathbb{R}^2 \setminus \overline{G}$ containing Q' . The whole closed cell Q' , being connected and disjoint from $\overline{G}$, lies in that same component.

If γ is an outer boundary curve of its physical component of G , assign Q to that outer curve. There are at most q such curves. If γ is an inner curve, its local complementary side is its bounded Jordan interior. The component C cannot cross γ , and hence is contained in that interior. In particular, the Jordan interior contains the whole empty h -cell Q' . Its boundary therefore has length at least $2h$: its horizontal span is at least h , and the two arcs joining a horizontal minimum to a maximum each have length at least that span. Nested components of G inside the hole do not change this argument. We may thus charge every occupied boundary cell to a curve which is either one of the q outer curves or an inner curve of length at least $2h$.

A closed rectifiable curve of length $\ell > 0$ meets at most

$$16 \left\lceil \frac{\ell}{h} \right\rceil \leq 16 \left(\frac{\ell}{h} + 1 \right)$$

closed h -cells. To verify the bound, partition it into $\lceil \ell/h \rceil$ arcs of length at most h . Each arc lies in the closed radius- h disk about its initial point. The projection of that disk on either coordinate axis has length $2h$ and meets at most four closed grid intervals, including possible endpoint contacts. The disk therefore meets at most sixteen closed cells.

The total length of the charged curves is at most p . There are at most $p/(2h)$ charged inner curves and at most q charged outer curves. Thus the number of occupied cells incident to an exposed edge is at most

$$16 \left(\frac{p}{h} + q + \frac{p}{2h} \right) = 24 \frac{p}{h} + 16q.$$

Each cell has four edges, proving (12.2). No charge for an arbitrarily large population of tiny holes remains: holes which fail to contain an empty grid cell are already filled by the selected mask.

Deleting whole open components of U cannot increase $b(U)$. Different open cell components share no full cell edge, so removing one exposes no previously internal edge of another. Translating the retained components into separate bounding boxes preserves their total exposed-edge count. Dilation to unit pitch changes lengths but not this count. Consequently (12.2) continues to bound the packed unit-cell witnesses used above.

12.3. Boundary budgets for the simultaneous library. In the construction of (8.3), the smooth active set satisfies

$$p \leq 3H, \quad q \leq \frac{(1+d/4)(H+r_2)}{2\pi} \leq \frac{25H}{32\pi} < H, \quad h = \frac{d}{100H},$$

where $H \geq 1$, $0 < d \leq 1$, and $r_2 \leq H/4$. Equation (12.2) therefore gives, for its packed witness P ,

$$b(P) \leq 28800 \frac{H^2}{d} + 64H \leq 28864 \frac{H^2}{d}. \quad (12.3)$$

The same witnesses as in (8.1) and (8.2), with no additional spectral loss, satisfy respectively

$$b(P) \leq \left\lceil 135300E^3\tau^{-1} \right\rceil, \quad 0 < \tau \leq 1, \quad (12.4)$$

$$b(P) \leq \left\lceil 180400E^2\varepsilon^{-1} \right\rceil, \quad 0 < \varepsilon \leq 1. \quad (12.5)$$

Indeed, in the additive case

$$\frac{H^2}{d} \leq \frac{25}{16}E^2\frac{3E}{\tau}, \quad 28864 \cdot \frac{75}{16} = 135300.$$

In the relative case

$$\frac{H^2}{d} \leq \frac{25}{16}E^2\frac{4}{\varepsilon}, \quad 28864 \cdot \frac{25}{4} = 180400.$$

For the exact-index library, $E = 20k$, so

$$b(P) \leq b_k(\varepsilon) := \left\lceil 72160000k^2\varepsilon^{-1} \right\rceil. \quad (12.6)$$

Filtering the cell library (8.4) by this boundary budget retains every witness needed for (8.5). More generally, the common-prefix libraries may be filtered by (12.4) or (12.5); the same mask continues to control every requested index.

12.4. The boundary budget in the area-rank construction. For clarity we also track the boundary cost of the scalar area-rank construction. Use a smooth unit-area source with $\lambda_k \leq E$, parameters $0 < \eta, d \leq 1$, and

$$c = \frac{k\eta}{4E^2}, \quad C = \frac{36}{7}.$$

Let A , $a = |A| > 0$, and r be supplied by (6.6). Apply (7.2) to A with spectral allowance $(d/4)(1+\eta)E$, and delete inactive components at $H = (1+d/4)(1+\eta)E$. The resulting smooth set satisfies

$$p \leq 2\sqrt{2/c}a, \quad q \leq \frac{(1+d/4)aH}{2\pi} \leq \frac{25aE}{16\pi} < aE.$$

At the pitch $h = d\sqrt{c}/(32C)$, the boundary estimate becomes

$$b(U) \leq 6144\sqrt{2}C \frac{a}{cd} + 64aE.$$

By (7.3), $E \geq 2\pi k$, so $Ec = k\eta/(4E) \leq \eta/(8\pi) < 1$. In particular $Ecd < 1$. Since $6144\sqrt{2}C + 64 < 48000$, we obtain

$$b(U) \leq 48000 \frac{a}{cd} \leq 192000 \frac{E^2}{k\eta d}. \quad (12.7)$$

Passing to the connected component selected for the scalar ratio cannot increase this count.

At the localized-source interface (11.9)–(11.11) of the all-index construction, $k \leq 96\pi\varepsilon^{-3}$ and $E/k < 8\pi$. For the choice $\eta = d = \varepsilon/10$ used there, (12.7) gives

$$b(P) \leq 11796480000\pi^3\varepsilon^{-5} < 2^{42}\varepsilon^{-5}.$$

The last inequality follows already from $\pi < 22/7$. Thus the all-index catalogue may simultaneously impose

$$b(P) \leq b_\varepsilon := \left\lceil 2^{42}\varepsilon^{-5} \right\rceil \quad (12.8)$$

along with the budgets (11.14), namely the cell budget $B_\varepsilon = \lceil 2^{46}\varepsilon^{-6} \rceil$ and index budget $J_\varepsilon = \lceil 302\varepsilon^{-3} \rceil$. The ratio witness and its error are unchanged.

12.5. Reconstruction from a set of boundary edges. Fix an integer cell budget $B \geq 1$. All packed masks under consideration fit in the rectangle of $2B$ columns and B rows. Its possible cell edges, including the exterior perimeter of the rectangle, number

$$M = 2B(B + 1) + B(2B + 1) = 4B^2 + 3B \leq 7B^2.$$

A mask determines the subset of edges adjacent to exactly one occupied cell. Conversely, this subset determines the mask uniquely when all exterior cells are declared empty.

Indeed, label every cell by zero or one according as it is empty or occupied. Across a listed edge the label changes, and across an unlisted edge it does not. Starting from the exterior label zero, propagate these requirements through the finite cell adjacency graph. Every cell is reachable from the exterior. Inconsistent requirements reject the edge subset; consistent requirements determine all labels uniquely. One can then check that the reconstructed subset has the required area, component count, and exposed edges. In particular, there is no additional choice of loop matching, nesting, or interior slit arrangement. The domain is the interior of the union of the reconstructed closed cells, with all common cell edges opened.

For an integer boundary budget $b \geq 1$, the number of subsets with at most b edges is bounded by

$$\sum_{i=0}^{\min\{b, M\}} \binom{M}{i} \leq (M + 1)^b.$$

To prove the bound without restrictions on the relative sizes of b and M , number the available edges from 1 to M . List the labels of a chosen subset in increasing order and pad with zeros to length b . This injects the subsets into $\{0, 1, \dots, M\}^b$, also when $b \geq M$. Consequently

$$\#\{\text{admissible geometric candidates}\} \leq (7B^2 + 1)^b. \quad (12.9)$$

Most encoded subsets need not be boundaries of valid masks; discarding them, requiring at most B occupied cells, or requiring a specified component bound can only reduce the count. Every witness constructed above survives the appropriate filters.

For the exact-index problem, combine (8.4), (12.6), and (12.9). Uniformly for $k \geq 1$ and $0 < \varepsilon \leq 1$,

$$\log \#\{\text{geometric candidates}\} = O\left(k^2 \varepsilon^{-1} \log \frac{2k}{\varepsilon}\right). \quad (12.10)$$

For the all-index problem, the geometric count has logarithm

$$O\left(\varepsilon^{-5} \log \frac{2}{\varepsilon}\right). \quad (12.11)$$

Including the possible indices multiplies the count by at most J_ε and leaves this logarithmic order unchanged. For the additive common-prefix library, the analogous bound is $O(E^3 \tau^{-1} \log(2E/\tau))$; inserting an objective-independent energy ceiling gives one geometric library before the objective is specified.

These estimates count geometric candidates and, when stated, domain-index pairs. Section 13 supplies the continuum spectral certificates to be applied to those candidates.

After canonical packing and removal of duplicates, the filtered library is also a subfamily of the cell-encoded library. One may therefore use the minimum of the cell-encoding and boundary-encoding counts. The boundary estimate improves the inverse-tolerance power in the logarithm, whereas the cell estimate can be smaller in other joint ranges of index and tolerance. Neither count is asserted optimal for a specially chosen spectrally dominating library. Geometric counting alone does not identify which different hole placements are redundant for the spectral objective.

13. UNIFORM CERTIFICATION BY SPECTRAL CLIPPING

The finite reductions become effective only after their spectral evaluations have a certified continuum meaning. We use a compression of the ambient Dirichlet Laplacian with its spectrum clipped at a finite height. The resulting lower hierarchy is determined by one finite Gram

matrix. Its error will be estimated through a local potential penalty, but the two operators are distinct; an exact compressed harmonic-mean identity connects them.

Square-root large-coupling estimates have substantial antecedents. Bruneau and Carbou obtain smooth boundary-layer and spectral expansions [7], and Agbanusi proves a compressed L^2 norm-resolvent estimate of square-root order for smooth transmission geometry [1, Theorem 3.1]. Verified polygonal spectra and certified finite-pixel computation are also established [22, 28]. Here the quantitative point is an explicit estimate for the *clipped* operator, uniform over the permitted mask topology, together with its finite-matrix realization.

13.1. Masks, form domains, and the finite hierarchy. Initially let $Q = (0, b)^2$, with b a positive integer. A square mask is the ordinary open interior of a finite nonempty union of closed axis-parallel unit lattice cells contained in $\overline{Q}$. Later we also allow the open interior of a selection from a supplied finite conforming triangulation of $\overline{Q}$. In either case the mesh covers the whole ambient square. The open set itself determines the Dirichlet condition: shared occupied edges are interior, while distinct occupied sectors may meet only at a boundary vertex. Thus holes, disconnection, and point contacts are allowed, without adding Dirichlet slits to the geometric representation.

Write $A_G = -\Delta_G$ for the Dirichlet operator on an open set G , and $J : L^2(P) \rightarrow L^2(Q)$ for zero extension. For either mask class the exact form-domain identity is

$$JH_0^1(P) = \{v \in H_0^1(Q) : v = 0 \text{ almost everywhere on } Q \setminus P\}. \quad (13.1)$$

The forward inclusion follows from compact smooth approximation. For the converse, extend v by zero outside Q and truncate its real and imaginary parts to bounded $H^1(\mathbb{R}^2)$ functions. These truncations converge in H^1 and still vanish outside P . For a bounded truncation remove every mesh vertex with a radial cutoff that is zero on the disk of radius ε^2 , one outside the disk of radius ε , and affine in $\log r$ between them. Its Dirichlet integral is $2\pi/\log(1/\varepsilon)$. There are only finitely many vertices, so their disks can be chosen disjoint and the product cutoff χ_ε satisfies

$$\int |v \nabla \chi_\varepsilon|^2 \leq \|v\|_\infty^2 \int |\nabla \chi_\varepsilon|^2 \rightarrow 0.$$

The other terms in $v\chi_\varepsilon - v$ tend to zero by absolute continuity of the integrals on shrinking disks. Away from these vertices, a boundary patch is a straight edge. The global H^1 function has equal traces on its two sides, and its empty-side trace is zero. Local zero-trace density on a half-plane, interior mollification, and a finite partition of unity therefore put $v\chi_\varepsilon$ in $H_0^1(P)$. Letting ε tend to zero and then removing the truncation proves (13.1). The capacity and trace facts used in this argument are standard [16]; the finite vertex removal is what permits point contacts.

The normalized ambient eigenpairs are

$$e_{pq}(x, y) = \frac{2}{b} \sin \frac{p\pi x}{b} \sin \frac{q\pi y}{b}, \quad \alpha_{pq} = \frac{\pi^2}{b^2}(p^2 + q^2), \quad p, q \geq 1.$$

For $H > 0$ define on $L^2(P)$

$$B_H = J^* \min(A_Q, H) J, \quad q_H(u) = \sum_{p,q} \min(\alpha_{pq}, H) |\langle Ju, e_{pq} \rangle|^2. \quad (13.2)$$

Let $\mathcal{I}_H = \{(p, q) : \alpha_{pq} < H\}$ and define $S_H : \mathbb{C}^{|\mathcal{I}_H|} \rightarrow L^2(P)$ by $S_H \mathbf{e}_{pq} = (H - \alpha_{pq})^{1/2} e_{pq}|_P$. Then

$$B_H = H I - S_H S_H^*, \quad M_H = S_H^* S_H. \quad (13.3)$$

If $\nu_j(M_H)$ are the decreasing eigenvalues of this finite positive matrix, continued by zero, its complete lower hierarchy is

$$\ell_j(H) = H - \nu_j(M_H) \leq \lambda_j(P), \quad j \geq 1. \quad (13.4)$$

This follows from equality of the nonzero spectra of $S_H S_H^*$ and $S_H^* S_H$, with multiplicities, and from $q_H(u) \leq \int_P |\nabla u|^2$ on $H_0^1(P)$. The continuation in (13.4) includes the infinite-dimensional plateau H .

The levels are increasing in H and converge to the Dirichlet levels. Here is a direct compactness argument identifying the limit. For $H > \lambda_j(P)$, the first j clipped levels lie below the plateau. Unit vectors in their joint subspace have clipped energy at most $\lambda_j(P)$. If α_{M+1} is the first ambient eigenvalue beyond a fixed finite list, then, for $H \geq \alpha_{M+1}$, their ambient Fourier tails have squared norm at most $\lambda_j(P)/\alpha_{M+1}$. Hence these unit balls are uniformly precompact in $L^2(Q)$. Along a sequence $H \rightarrow \infty$, orthonormal bases have j orthonormal strong limits supported in P . Passing first to each fixed bounded clipped form and then increasing its cutoff shows that the limiting subspace has finite Dirichlet energy and Rayleigh quotient at most $\lim_H \ell_j(H)$. Identity (13.1) puts it in $H_0^1(P)$. Min-max gives

$$\ell_j(H) \uparrow \lambda_j(P). \quad (13.5)$$

For arithmetic, choose an integer c and $H = \pi^2 c/b^2$. The active set is exactly $\mathcal{I}_c = \{(p, q) \in \mathbb{N}^2 : p^2 + q^2 < c\}$. With

$$G_{ab} = \int_P e_a e_b, \quad w_a = c - p_a^2 - q_a^2, \quad T_c = \text{diag}(\sqrt{w}) G \text{diag}(\sqrt{w}),$$

we have $0 \leq G \leq I$ and

$$\ell_j(c) = \frac{\pi^2}{b^2} (c - \nu_j(T_c)). \quad (13.6)$$

Modes at equality have zero correction weight and are omitted. In particular, rounding π never determines cutoff membership. The disjoint squares $[p-1, p] \times [q-1, q]$ associated with the active indices fit inside the quadrant disk of radius $\sqrt{c}$, so the matrix dimension satisfies

$$N_c = |\mathcal{I}_c| \leq \frac{\pi c}{4} < c. \quad (13.7)$$

13.2. A boundary identity with corners and point contacts. The quantitative argument uses the normal derivative of $u = (A_P + s)^{-1}f$, where $s > 0$ and $f \in L^2(P)$. Let Γ be the union of the open exposed edges. The following qualitative justification will be used for both meshes. Near a boundary vertex, treat each locally connected occupied sector separately. Its angle ω is strictly between 0 and 2π . For a nonflat sector, the local Dirichlet expansion for an L^2 right-hand side has an H^2 regular part w with $r^{-1}\nabla w \in L^2$, and, when $\pi < \omega < 2\pi$, one additional term

$$a r^\beta \sin(\beta\varphi), \quad \beta = \pi/\omega > 1/2.$$

This is the sector statement in [14, Theorem L.1]; localization is legitimate because the cutoff commutator contains only u and ∇u . Flat sectors are treated by reflection. In particular, $g = \partial_n u$ belongs to $L^2(\Gamma)$: the singular normal derivative has square integrable size $r^{\beta-1}$.

For a bounded, cellwise smooth vector field $h \in W^{1,\infty}(P)$, remove small disks at the finitely many nonflat boundary sectors and integrate cell by cell. Internal-edge terms cancel because h has a common trace and u is locally H^2 across an interior edge. The artificial-circle terms tend to zero along a common sequence of radii. Indeed,

$$\int_{\varepsilon < r < 2\varepsilon} |\nabla w|^2 \leq 4\varepsilon^2 \int_{\varepsilon < r < 2\varepsilon} r^{-2} |\nabla w|^2 = o(\varepsilon^2).$$

Sum this estimate over the finite sector set before averaging in r . One obtains a common radius in $[\varepsilon, 2\varepsilon]$ where the sum of regular-part circle integrals is $o(\varepsilon)$. The singular-part circle integrals are $O(r^{2\beta-1})$, and hence also vanish.

If $h \cdot n = 1$ on Γ , applying this argument to $h|\nabla u|^2 - 2\text{Re}((h \cdot \nabla \bar{u})\nabla u)$ gives the Rellich identity

$$\begin{aligned} \int_\Gamma |\partial_n u|^2 &= 2\text{Re} \int_P (\Delta u)(h \cdot \nabla \bar{u}) \\ &\quad + 2\text{Re} \int_P \sum_{i,k} (\partial_i h_k)(\partial_k \bar{u})(\partial_i u) - \int_P (\text{div } h)|\nabla u|^2. \end{aligned} \quad (13.8)$$

On an exposed edge the tangential derivative is zero, giving the boundary term $-|\partial_n u|^2$ before rearrangement. The same puncture argument proves Green's identity against smooth ambient

tests: Cauchy–Schwarz contributes the additional circle-length factor to the normal boundary integral. The exterior trace bounds proved below then extend that identity to every test in $H_0^1(Q)$. Thus only qualitative corner regularity is used; no corner-dependent regularity constant enters the numerical bounds.

13.3. Uniform flux and exterior trace on unit masks. Construct h by bilinear interpolation on each occupied unit cell. At a lattice vertex assign its value separately on each locally connected occupied sector. For one or three occupied quadrants, take the sum of the two distinct exposed outward unit normals. For two adjacent quadrants take their common outward normal; for a full vertex take zero. Two opposite quadrants are treated separately.

Occupied cells sharing an edge have the same endpoint values and hence the same interpolated trace there. Distinct values at a point contact cause no jump inside P , even if the sectors reconnect elsewhere. The nodal components belong to $\{-1, 0, 1\}$, and bilinear interpolation therefore gives

$$|h| \leq \sqrt{2}, \quad |\partial_i h_k| \leq 2, \quad \|Dh\|_{\text{op}} \leq 4, \quad |\operatorname{div} h| \leq 4, \quad h \cdot n = 1 \text{ on } \Gamma. \quad (13.9)$$

Every estimate is local and independent of the number of cells. The true sector angles are $\pi/2$ and $3\pi/2$, so the argument of Section 13.2 applies. By spectral calculus,

$$\|\Delta u\|_2 \leq \|f\|_2, \quad \|\nabla u\|_2 \leq \sup_{t \geq 0} \frac{\sqrt{t}}{t+s} \|f\|_2 = \frac{1}{2\sqrt{s}} \|f\|_2.$$

Consequently (13.8) yields, for $s \geq 1$,

$$\begin{aligned} \|\partial_n(A_P + s)^{-1} f\|_{L^2(\Gamma)}^2 &\leq 2\sqrt{2} \|\Delta u\|_2 \|\nabla u\|_2 + 12 \|\nabla u\|_2^2 \\ &\leq (\sqrt{2}s^{-1/2} + 3s^{-1}) \|f\|_2^2 \leq 5s^{-1/2} \|f\|_2^2. \end{aligned} \quad (13.10)$$

For $v \in H_0^1(Q)$, extended by zero to the plane, put $M = \int_{P^c} |v|^2$ and $T = \int_{P^c} |\nabla v|^2$. Integration of $((1-t)|f(t)|^2)'$ for $f \in H^1(0, 1)$ gives

$$|f(0)|^2 \leq \int_0^1 |f|^2 + 2 \left(\int_0^1 |f|^2 \int_0^1 |f'|^2 \right)^{1/2}.$$

Apply this along the normals through the empty unit cell adjacent to each exposed edge. An empty cell is charged at most four times for its mass and twice for each directional derivative. Cauchy–Schwarz over the edges proves

$$\|\gamma v\|_{L^2(\Gamma)}^2 \leq 4M + 4\sqrt{2}\sqrt{MT}. \quad (13.11)$$

The same argument is valid for adjacent cells crossing or lying outside Q , using zero extension; exposed edges on ∂Q have zero trace. If

$$a_{\kappa,s}(v) = \int_Q |\nabla v|^2 + \kappa \int_{Q \setminus P} |v|^2 + s \|v\|_2^2, \quad v \in H_0^1(Q), \quad (13.12)$$

then $M \leq \kappa^{-1} a_{\kappa,s}(v)$ and $T \leq a_{\kappa,s}(v)$. Thus, for $\kappa \geq 1$ and $s > 0$,

$$\|\gamma v\|_{L^2(\Gamma)}^2 \leq 12\kappa^{-1/2} a_{\kappa,s}(v). \quad (13.13)$$

13.4. The positive residual and the exact clipping comparison. Let $K_\kappa = A_Q + \kappa 1_{Q \setminus P}$ be the operator of the unshifted part of (13.12). We first record the argument with two geometric constants. Suppose, for $s, \kappa \geq 1$, that

$$\|\partial_n(A_P + s)^{-1} f\|_{L^2(\Gamma)}^2 \leq F s^{-1/2} \|f\|_2^2, \quad \|\gamma v\|_{L^2(\Gamma)}^2 \leq C \kappa^{-1/2} a_{\kappa,s}(v). \quad (13.14)$$

Unit masks have $F = 5$ and $C = 12$.

For arbitrary $f \in L^2(Q)$ set

$$u_\kappa = (K_\kappa + s)^{-1} f, \quad u_D = J(A_P + s)^{-1} J^* f, \quad e = u_\kappa - u_D.$$

The form restricted to $JH_0^1(P)$ is the shifted Dirichlet form. Hence $a_{\kappa,s}(e, w) = 0$ for every such w , and

$$\left\langle ((K_\kappa + s)^{-1} - J(A_P + s)^{-1}J^*)f, f \right\rangle = a_{\kappa,s}(e, e) \geq 0. \quad (13.15)$$

Indeed, expand $u_\kappa = u_D + e$ and use orthogonality, or test the variational equation by e . Write $f_P = J^*f$ and $f_{\text{ext}} = f|_{Q \setminus P}$. Green's identity gives

$$a_{\kappa,s}(e, v) = \int_{Q \setminus P} f_{\text{ext}} \bar{v} - \int_\Gamma \partial_n u_D \bar{\gamma} v.$$

By (13.14) its absolute value is at most

$$\left(\kappa^{-1/2} \|f_{\text{ext}}\|_2 + \sqrt{FC} s^{-1/4} \kappa^{-1/4} \|f_P\|_2 \right) a_{\kappa,s}(v)^{1/2}.$$

The two pieces of f are orthogonal. Squaring by Cauchy-Schwarz on these two components, taking $v = e$, and using (13.15) proves the full ambient operator inequality

$$0 \leq (K_\kappa + s)^{-1} - J(A_P + s)^{-1}J^* \leq \left(\frac{FC}{\sqrt{s\kappa}} + \frac{1}{\kappa} \right) I. \quad (13.16)$$

The positive quadratic identity is essential: the form-dual residual has quarter-power size, and its square gives the resolvent estimate.

To compare with clipping, put $T = A_Q + s$ and $\kappa = H + s$. The exact scalar shift identity and the harmonic-mean inequality give

$$B_H + s = J^* \min(T, \kappa) J \geq J^* \frac{\kappa T}{T + \kappa} J.$$

Let $X = J^*(T + \kappa)^{-1}J$. Since $\|\kappa X\| \leq \kappa/(s + \kappa) < 1$, all the following bounded inverses exist. The resolvent identity for the bounded perturbation $-\kappa J J^*$ of $T + \kappa$ gives

$$J^*(K_\kappa + s)^{-1}J = X(I - \kappa X)^{-1}.$$

The compressed harmonic mean is $\kappa(I - \kappa X)$, and therefore

$$\left[J^* \frac{\kappa T}{T + \kappa} J \right]^{-1} = \kappa^{-1} I + J^*(K_\kappa + s)^{-1}J. \quad (13.17)$$

Inversion reverses positive operator order. Conversely, $q_H \leq \int_P |\nabla u|^2$ on $H_0^1(P)$ and the variational inverse formula $\langle A^{-1}f, f \rangle = \sup_v (2 \operatorname{Re} \langle f, v \rangle - a(v))$ give $(B_H + s)^{-1} \geq (A_P + s)^{-1}$. Combining these facts with (13.16) yields, for $s \geq 1$ and $H > 0$,

$$0 \leq (B_H + s)^{-1} - (A_P + s)^{-1} \leq \left(\frac{FC}{\sqrt{s(H + s)}} + \frac{2}{H + s} \right) I \leq \frac{FC + 2}{\sqrt{s(H + s)}} I. \quad (13.18)$$

In particular the proof uses neither an exchange of compression and inversion nor an ordering between the eigenvalues of the local penalty and those of B_H .

The decreasing variational levels of $(B_H + s)^{-1}$ are $(\ell_j(H) + s)^{-1}$, continued at $(H + s)^{-1}$. The bounded-operator min-max principle applies to (13.18), although this resolvent is not compact. Whenever $\lambda_j(P) \leq E$, taking $s = E + 1$ gives

$$0 \leq \lambda_j(P) - \ell_j(H) \leq 4(FC + 2)(E + 1)^{3/2} H^{-1/2}. \quad (13.19)$$

Here both denominators used to pass from reciprocal levels to levels are at most $E + s \leq 2s$. For unit masks $FC = 60$. Rounding 248 upwards gives the convenient statement

$$\boxed{0 \leq \lambda_j(P) - \ell_j(H) \leq 256(E + 1)^{3/2} H^{-1/2}, \quad 0 < \lambda_j(P) \leq E, \quad H > 0.} \quad (13.20)$$

The constants are independent of cell count, holes, components, point contacts, and ambient side length. The physical unit-cell scale has been fixed.

13.5. Cutoff, normalization, and sharpness. For $E \geq 1$ and $0 < \tau \leq 1$, the explicit choice

$$H_*(E, \tau) = 2^{18}(E+1)^3\tau^{-2} \quad (13.21)$$

makes the continuum error at most $\tau/2$. Arithmetic intervals of width at most $\tau/2$ for the clipped levels then give continuum intervals of width at most τ ; conforming upper bounds are optional improvements, not an additional convergence assumption.

For the catalogues, let the raw mask have $n_P \leq B$ unit cells and use its packed ambient square of side $b = 2B$. Its unit-area normalization is $P^\# = n_P^{-1/2}P$, with $\lambda_j(P^\#) = n_P\lambda_j(P)$. Thus a common raw tolerance τ/B suffices. Substitution in (13.21), using $\pi^2 > 9$, gives the integer mode cutoff

$$c = \left\lceil \frac{2^{20}}{9} B^4 (E+1)^3 \tau^{-2} \right\rceil, \quad N_c \leq \frac{\pi c}{4}. \quad (13.22)$$

Here E bounds the *raw unit-cell* eigenvalues. Every nonempty mask contains an open unit square. If $a = \lceil \sqrt{m} \rceil$ and $d = \lceil m/a \rceil$, then $ad \geq m$, $d \in \{a-1, a\}$, and $a^2 + d^2 \leq 3m$. For $d = a$, use $m \geq a(a-1) + 1$; for $d = a-1$, use $m \geq (a-1)^2 + 1$; $m = 1$ is immediate. The corresponding rectangular block of square modes and domain monotonicity give

$$\lambda_m(P) \leq 3\pi^2 m < 30m. \quad (13.23)$$

Thus $E = 30m$ yields a matrix dimension $O(B^4 m^3 \tau^{-2})$ for every normalized prefix through m . The bit-cost argument in Appendix A is polynomial in B, m, τ^{-1} . This per-mask improvement leaves the outer geometric enumeration orders unchanged.

The power of H in (13.20) cannot be improved, even on one fixed mask. Take $Q = (0, 3)^2$ and $P = (1, 2) \times (0, 3)$, the union of three unit cells. Its normalized first eigenfunction, extended by zero, is

$$u(x, y) = \sqrt{2} \sin(\pi(x-1)) \sqrt{\frac{2}{3}} \sin \frac{\pi y}{3}, \quad \lambda_1(P) = \frac{10\pi^2}{9} < 12.$$

Only the ambient y mode 1 occurs. Direct sine integration gives, for the ambient x coefficient and $m \neq 3$,

$$c_m = -\frac{2\pi}{\sqrt{3}} \frac{\sin(m\pi/3) + \sin(2m\pi/3)}{(m\pi/3)^2 - \pi^2}.$$

Hence $|c_m| = 18/(\pi|m^2 - 9|)$ when $m \equiv 1, 5 \pmod{6}$, with all other nonresonant coefficients zero. Writing $\alpha_m = \pi^2(m^2 + 1)/9$, the Rayleigh principle gives

$$\lambda_1(P) - \ell_1(H) \geq \lambda_1(P) - q_H(u) = \sum_{\alpha_m > H} (\alpha_m - H) |c_m|^2.$$

For $H \geq 36$ put $M = \lceil 2\sqrt{H} \rceil$. Then $M \geq 12$, $M \leq 3\sqrt{H}$, and $\alpha_m > 2H$ on $[M, 2M]$. Each eligible residue in that interval contributes at least

$$\frac{\pi^2 m^2}{18} \frac{324}{\pi^2 m^4} \geq \frac{18}{4M^2}.$$

There are at least $M/3 - 2 \geq M/6$ eligible integers. Therefore

$$\lambda_1(P) - \ell_1(H) \geq \frac{3}{4M} \geq \frac{1}{4\sqrt{H}}, \quad H \geq 36. \quad (13.24)$$

This proves sharpness of the clipping decay exponent; it does not determine the optimal numerical coefficient or energy power in (13.20).

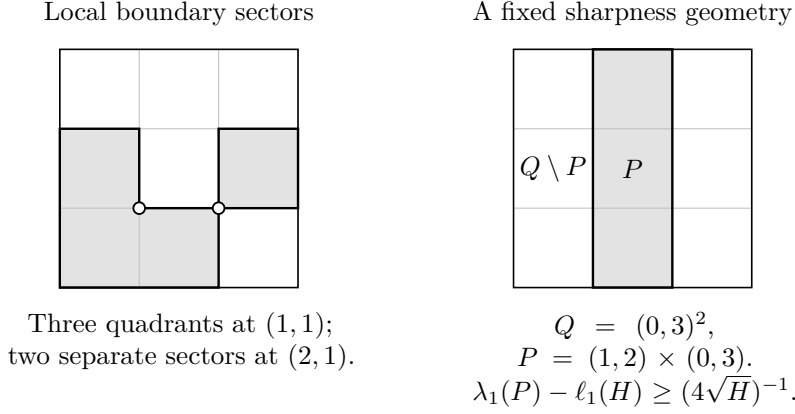


FIGURE 2. Clipping certificates use the actual open interior of the shaded cells. Left: the bilinear boundary field has value $(1, 1)$ at the reentrant vertex $(1, 1)$, but has two independent sector values $(1, 1)$ and $(-1, -1)$ at the point contact $(2, 1)$. These point values do not join the open components. Right: the rectangular mask used in (13.24) shows that the $H^{-1/2}$ clipping rate is already necessary on one fixed geometry; the displayed lower bound holds for $H \geq 36$.

13.6. A supplied conforming triangular mesh. Let a finite conforming straight-sided triangulation cover $\overline{Q}$, and let P be the interior of a nonempty union of its selected closed triangles. Assume every angle of every ambient triangle is at least $\theta > 0$ and every altitude is at least $\sigma > 0$. These bounds include the unselected triangles. Set

$$A_\theta = \csc(\theta/2), \quad B_\theta = \csc^2 \theta, \quad D_\theta = 4\{24A_\theta(1 + B_\theta) + 2\}. \quad (13.25)$$

We claim that

$$0 \leq \lambda_j(P) - \ell_j(H) \leq D_\theta(E + \sigma^{-2})^{3/2}H^{-1/2}, \quad \lambda_j(P) \leq E, \quad H > 0. \quad (13.26)$$

First rescale so that every altitude is at least one. At each boundary vertex assign a nodal value separately on each occupied sector of opening ω . For $\omega \neq \pi$ solve $h \cdot n_1 = h \cdot n_2 = 1$ for the two outward boundary normals; its norm is $1/\sin(\omega/2)$. For a flat sector use the common normal, and at a full interior vertex use zero. Every nonempty occupied sector and every missing sector at an interior boundary vertex contains a triangle, so $\theta \leq \omega \leq 2\pi - \theta$. At an ambient boundary vertex the exterior of the square supplies the missing angle. Thus all nodal values have norm at most A_θ .

Interpolate affinely on each occupied triangle. Across a shared occupied edge the endpoint values agree, while contact sectors are again independent. Barycentric basis gradients have norm at most one, the reciprocal of the respective altitudes. Consequently

$$|h| \leq A_\theta, \quad \|Dh\|_{\text{op}} \leq 3A_\theta, \quad |\operatorname{div} h| \leq 6A_\theta, \quad h \cdot n = 1.$$

The corner exponent is $\beta = \pi/\omega \geq \pi/(2\pi - \theta) > 1/2$, so Section 13.2 justifies (13.8). For $s \geq 1$ it follows that

$$\|\partial_n(A_P + s)^{-1}f\|_{L^2(\Gamma)}^2 \leq A_\theta(s^{-1/2} + 3s^{-1})\|f\|_2^2 \leq 4A_\theta s^{-1/2}\|f\|_2^2. \quad (13.27)$$

For the exterior trace let e be an exposed interior edge, T_e its adjacent empty triangle, z the opposite vertex, and a_e the altitude to e . If L_{T_e} is the longest side, then

$$L_{T_e}/a_e \leq \csc^2 \theta = B_\theta.$$

Indeed, the sine law gives every side length at least $L_{T_e} \sin \theta$, and the altitude is such a side times the sine of a base angle, again at least $\sin \theta$. The vector field $X_e(x) = (x - z)/a_e$ on T_e has outward normal component one on e and zero on its other two edges, with $\operatorname{div} X_e = 2/a_e \leq 2$

and $|X_e| \leq B_\theta$. The divergence theorem for $|v|^2 X_e$ therefore gives

$$\int_e |v|^2 \leq 2 \int_{T_e} |v|^2 + 2B_\theta \left(\int_{T_e} |v|^2 \int_{T_e} |\nabla v|^2 \right)^{1/2}.$$

Each empty triangle is charged at most three times. With $M = \int_{Q \setminus P} |v|^2$ and $T = \int_{Q \setminus P} |\nabla v|^2$ this yields

$$\|\gamma v\|_{L^2(\Gamma)}^2 \leq 6M + 6B_\theta \sqrt{MT} \leq 6(1 + B_\theta) \kappa^{-1/2} a_{\kappa,s}(v), \quad \kappa \geq 1.$$

Edges on ∂Q have zero trace. Thus (13.14) holds with $F = 4A_\theta$ and $C = 6(1 + B_\theta)$. Applying (13.19) proves (13.26) in the normalized geometry.

To restore physical units put $x = \sigma y$. The transformed energies are $E' = \sigma^2 E$ and $H' = \sigma^2 H$, and the transformed eigenvalue difference is $\sigma^2(\lambda_j - \ell_j)$. Hence

$$\lambda_j - \ell_j \leq D_\theta \sigma^{-2} (\sigma^2 E + 1)^{3/2} (\sigma^2 H)^{-1/2} = D_\theta (E + \sigma^{-2})^{3/2} H^{-1/2},$$

as asserted. A sufficient cutoff for error at most $\tau/2$ is

$$H \geq 4D_\theta^2 (E + \sigma^{-2})^3 \tau^{-2}. \quad (13.28)$$

The rectangle in (13.24) has a fixed right-triangle mesh, so the same H exponent is sharp in this larger class. For rational vertices the complete arithmetic construction is given in Appendix A.3; the constants retain their explicit dependence on the supplied mesh quality.

13.7. The finite matrix as a whole-resolvent approximation. The clipped inverse has an identity plateau and is not compact. The correct finite-rank approximation is obtained by subtracting exactly that plateau. Put $\kappa = H + s$, $R_{P,s} = (A_P + s)^{-1}$, and

$$F_{H,s} = (B_H + s)^{-1} - \kappa^{-1} I = \kappa^{-1} S_H (\kappa I - M_H)^{-1} S_H^*. \quad (13.29)$$

The last formula follows from (13.3), or by applying the scalar identity on each singular direction of S_H . Thus $F_{H,s}$ is positive and finite rank, determined by the same Gram matrix and the known restricted ambient modes.

For unit masks, $s \geq 1$ and $H > 0$, define

$$\eta_{H,s} = \frac{60}{\sqrt{s(H+s)}} + \frac{1}{H+s}. \quad (13.30)$$

The harmonic-mean comparison gives

$$0 \leq F_{H,s} \leq J^*(K_\kappa + s)^{-1} J \leq J^*(A_Q + s)^{-1} J.$$

Together with $(B_H + s)^{-1} \geq R_{P,s}$ and (13.16), this yields

$$-\kappa^{-1} I \leq F_{H,s} - R_{P,s} \leq \eta_{H,s} I, \quad \|F_{H,s} - R_{P,s}\|_{\text{op}} \leq \eta_{H,s}.$$

For a planar finite-area Dirichlet domain G , free heat domination gives

$$N_{(A_G+s)^{-1}}(t) \leq N_{A_G}(1/t) \leq e \operatorname{Tr}(e^{-tA_G}) \leq \frac{e}{4\pi} |G| t^{-1}, \quad t > 0,$$

where the left count uses eigenvalues strictly greater than t and the middle count uses energies below the stated threshold. Compression does not increase ordered positive eigenvalues. For the compact self-adjoint error $D = F_{H,s} - R_{P,s}$, its positive eigenvalues are bounded by those of $F_{H,s}$ and its negative magnitudes by those of $R_{P,s}$. Therefore

$$N_{|D|}(t) \leq \frac{e}{4\pi} (|P| + |Q|) t^{-1}, \quad N_{|D|}(t) = 0 \quad (t > \eta_{H,s}).$$

Integration of this count proves, for every $p > 1$,

$$\|F_{H,s} - R_{P,s}\|_{\mathcal{S}_p}^p \leq \frac{p}{p-1} \frac{e}{4\pi} (|P| + |Q|) \eta_{H,s}^{p-1}. \quad (13.31)$$

Thus the finite hierarchy approximates the entire shifted resolvent in every planar Schatten class above its summability threshold. For nonempty P the threshold is strict: an interior square already forces the Dirichlet inverse to lie outside $\mathcal{S}_p$ for $0 < p \leq 1$, and a finite-rank

perturbation cannot remove this tail. Without plateau subtraction the error is $\kappa^{-1}I$ plus a compact operator, hence is noncompact for every finite H .

For triangular masks the same assertions hold for $s \geq \sigma^{-2}$ after replacing (13.30) by

$$\eta_{H,s;\theta,\sigma} = \frac{24 \csc(\theta/2)(1 + \csc^2 \theta)}{\sqrt{s(H+s)}} + \frac{1}{H+s}.$$

Under $x = \sigma y$, the normalized resolvent is multiplied by σ^2 on returning to physical units, while H and s both acquire the factor σ^2 . These factors give exactly the displayed expression; the areas in (13.31) are physical areas. The Schatten rate here follows from the counting bound and is not asserted to be sharp.

14. EFFECTIVE OPTIMIZATION AND FINITE CHECKS

The preceding constructions separate three costs: the size of the geometric library, the cutoff needed for each continuum spectrum, and the arithmetic precision of its finite matrix. We now combine them into value enclosures and witnesses. No comparison of two exact eigenvalues is required.

For the exact-index problem, let a dyadic tolerance $0 < \varepsilon \leq 1$ be given. Use the boundary-filtered library of (8.4) and (12.6) with $\gamma = \varepsilon/4$. Write M for its true minimum. Enclose each candidate's normalized k -th eigenvalue in a rational interval of width at most $\eta = \varepsilon/4$, using Section 13 and Appendix A. The minima l, u of the lower and upper endpoints satisfy

$$l \leq M \leq u, \quad u - l \leq \eta.$$

For the width assertion, use the candidate attaining l : its upper endpoint is at most $l + \eta$, and u is no larger. Equation (8.5) gives the fully rational enclosure

$$\mu_k \in \left[\frac{l}{1+\gamma}, u \right]. \quad (14.1)$$

Its lower endpoint is positive. Indeed, $l \geq M - \eta \geq 2\pi k - \eta > 23/4$. The ratio of the upper endpoint to the lower endpoint is at most

$$(1+\gamma)\frac{u}{l} \leq 1 + \gamma + (1+\gamma)\frac{\eta}{l} < 1 + \left(\frac{1}{4} + \frac{5}{92}\right)\varepsilon < 1 + \varepsilon.$$

Thus this is a relative enclosure with the requested accuracy, including all multiplicities and without assuming that the domain infimum is attained. A candidate attaining the least upper endpoint also satisfies

$$\lambda_k(P_*^\#) \leq u \leq M + \eta \leq (1+\gamma)\mu_k + \eta < (1+\varepsilon)\mu_k,$$

so the computation returns an actual approximate minimizing domain.

For the all-index problem use the connected, boundary-filtered catalogue with $\gamma = \varepsilon/64$ and its index budget J_γ . Enclose each ratio $|P|\lambda_j(P)/j$ to width at most $\eta = \varepsilon/4$. Enclosing the corresponding normalized eigenvalue to width η is sufficient since $j \geq 1$. The same minimum-endpoint construction yields

$$L \in \left[\frac{l}{1+\gamma}, u \right]. \quad (14.2)$$

Here $M \leq (1+\gamma)L$, so

$$u - \frac{l}{1+\gamma} \leq \eta + \frac{\gamma l}{1+\gamma} \leq \eta + \gamma L \leq \frac{\varepsilon}{4} + \frac{4\pi\varepsilon}{64} < \frac{\varepsilon}{2}.$$

A least-upper-endpoint candidate is within the same additive allowance of L . In particular, (14.2) computes the real number L to arbitrary absolute accuracy.

Boundary-edge reconstruction and validity tests use polynomial time in the containing grid dimensions. The proof of Appendix A gives polynomial per-mask bit cost in B, m, τ^{-1} , with all

continuum and arithmetic errors included. Substituting the catalogue budgets and multiplying by their candidate counts gives deterministic bit-cost upper bounds

$$\exp\left(O\left(k^2\varepsilon^{-1}\log\frac{2k}{\varepsilon}\right)\right), \quad \exp\left(O\left(\varepsilon^{-5}\log\frac{2}{\varepsilon}\right)\right) \quad (14.3)$$

for (14.1) and (14.2), respectively. The constants are absolute for these scalar tasks. The second search includes its J_γ indices; their polynomial factor is absorbed by the displayed bound. These estimates use a binary encoding of k and a dyadic accuracy request. An arbitrary rational input tolerance can first be rounded down to a dyadic number within a factor two, charging the input's bit length separately. For a general computable objective, the additional cost of its supplied evaluation algorithm remains a separate parameter, as in Section 9.

APPENDIX A. CERTIFIED ARITHMETIC AND PER-MASK BIT BOUNDS

The continuum estimates in Section 13 give an a-priori stopping cutoff. This appendix specifies the arithmetic needed to turn that cutoff into rational intervals. For packed unit-cell masks it also proves polynomial per-mask bit cost in the numerical parameters B, m, τ^{-1} . For a supplied rational triangular mesh the conclusion is termination with an explicit dimension and error budget.

A.1. Unit-cell Gram entries and arithmetic error. Let a nonempty raw mask P have $n_P \leq B$ cells in the packed square $Q = (0, b)^2$, $b = 2B$. Fix a prefix length $m \geq 1$ and a dyadic tolerance $0 < \tau \leq 1$ for its unit-area normalization. Put

$$t = \tau/B, \quad E = 30m, \quad c = \left\lceil \frac{2^{20}}{9} B^4 (E+1)^3 \tau^{-2} \right\rceil, \quad f = \frac{\pi^2}{b^2}. \quad (A.1)$$

By (13.22), the raw continuum error is at most $t/2$. The correction matrix T_c from (13.6) has dimension $N_c < c = O(B^4 m^3 \tau^{-2})$ and satisfies $0 \leq T_c \leq cI$.

In normalized square coordinates, define

$$J_{pr}(a, d) = 2 \int_a^d \sin(p\pi x) \sin(r\pi x) \, dx.$$

For $p \neq r$ this is

$$\left[\frac{\sin((p-r)\pi x)}{(p-r)\pi} - \frac{\sin((p+r)\pi x)}{(p+r)\pi} \right]_a^d, \quad (A.2)$$

whereas for $p = r$ it is

$$(d-a) - \frac{\sin(2p\pi d) - \sin(2p\pi a)}{2p\pi}.$$

Each Gram entry $G_{(p,q),(r,s)}$ is the sum over occupied cells of the corresponding products of two such integrals. The endpoints are rational. The separate diagonal expression handles repeated frequencies exactly.

For reference, an elementary rational enclosure for π is provided by

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239). \quad (A.3)$$

The identity follows from the tangent addition formula: $\tan(4 \arctan(1/5)) = 120/119$ and $\tan(4 \arctan(1/5) - \arctan(1/239)) = 1$, with the latter angle in $(0, \pi/2)$. Alternating arctangent series have next-term remainder bounds. Before evaluating a sine, reduce its rational multiple of π modulo two and use reflection to put the argument in $[-\pi/2, \pi/2]$. The sine Taylor polynomial with its Lagrange remainder then gives rational outward intervals. Integer square roots or rational bisection enclose each $\sqrt{w_a}$.

Assemble outward intervals for T_c , preserve symmetry, and round their midpoints to one common dyadic denominator, including that rounding in the radii. If A is the resulting rational symmetric midpoint matrix and r_{ab} are its entry radii, then

$$\|T_c - A\|_{\text{op}} \leq \Delta := \max_a \sum_b r_{ab}. \quad (A.4)$$

The matrix dimension is thus explicitly included in the rounding budget.

Here is one concrete allocation of the raw tolerance t . Choose a positive rational interval $[f_-, f_+]$ containing f , and refine the matrix and scalar intervals until

$$\Delta \leq \frac{t}{16f_+}, \quad f_+ - f_- \leq \frac{t}{8c}. \quad (\text{A.5})$$

Enclose each required decreasing eigenvalue of A in a rational interval of width $\zeta \leq t/(8f_+)$. Enlarging by Δ on each side gives an interval for $\nu_j(T_c)$; intersect it with $[0, c]$. The width is at most $\zeta + 2\Delta$. Inserting this interval and $[f_-, f_+]$ in (13.6) gives a rational interval $[L_j, U_j]$ for $\ell_j(c)$, of width at most

$$f_+(\zeta + 2\Delta) + c(f_+ - f_-) \leq \frac{t}{4} + \frac{t}{8} = \frac{3t}{8}.$$

Zero continuation is used if the index exceeds N_c . Adding the one-sided continuum allowance gives

$$\lambda_j(P^\#) \in [n_P L_j, n_P(U_j + t/2)], \quad 1 \leq j \leq m, \quad (\text{A.6})$$

with width at most $7\tau/8 < \tau$. The use of 2Δ , rather than Δ , in this interval width accounts for the possible overhang of an outward endpoint.

A.2. Exact inertia and polynomial bit cost. We next give a deterministic eigenvalue procedure, so that no success assumption about floating-point eigenvectors enters (A.6). For a rational symmetric matrix A and a dyadic shift a , compute the inertia of $A - aI$ by symmetric rational elimination. A nonzero diagonal entry is used as a one-by-one pivot. If every diagonal is zero but the matrix is not zero, a nonzero off-diagonal entry gives, after permutation, the invertible pivot

$$C = \begin{pmatrix} 0 & b \\ b & 0 \end{pmatrix},$$

whose inertia has one positive and one negative direction. If the remaining matrix is zero, its entire dimension is nullity. At each nonzero pivot,

$$\begin{pmatrix} C & F \\ F^T & G \end{pmatrix} \text{ is congruent to } \begin{pmatrix} C & 0 \\ 0 & G - F^T C^{-1} F \end{pmatrix}.$$

Congruence is an invertible change of coordinates in a real quadratic form and preserves inertia. The pivot inertia therefore adds to that of the smaller Schur complement. There are at most N_c stages and $O(N_c^3)$ rational operations.

To control bit growth, clear the common dyadic denominator of the original shifted matrix and suppose its integer entries have at most L_0 bits. Each later Schur-complement entry is a bordered minor divided by the determinant of the previously selected nonsingular principal block, with the initial scaling restored. This follows directly from the block determinant formula; after either pivot size, the selected original principal block remains nonsingular because its determinant is the product of the invertible pivot determinants. Hadamard's inequality bounds every determinant of order at most N_c by

$$N_c^{N_c/2} 2^{N_c L_0}.$$

Its binary length is $O(N_c(L_0 + \log(N_c + 1)))$. Reducing intermediate fractions by their integer greatest common divisors therefore keeps all stored entries at polynomial bit length. The products and sums in one elimination step also have polynomial bit length. An arbitrarily small nonzero real pivot creates no difficulty: its reciprocal remains a ratio of integers with the stated bit bounds.

A rational row-sum bound gives a dyadic interval $[-R, R]$ containing the spectrum of A . At a midpoint a , inertia gives the number of eigenvalues at most a as the negative count plus the nullity of $A - aI$. For the j th increasing eigenvalue, retain the upper half of its bracket if that

count is less than j , and the lower half otherwise. This remains valid when a is exactly an eigenvalue. After

$$O\left(1 + \log \frac{R+1}{\zeta}\right)$$

inertia computations the bracket has width at most ζ . Decreasing indices are obtained by reversing the order. The method counts multiplicities and requires no lower bound on a spectral gap. Computing all N_c indices separately still has polynomial bit cost.

For completeness, the construction of A also has the required polynomial bound. Cell endpoints and active mode indices have $O(\log(B+1) + \log(c+1))$ bits. The reduction of sine arguments is exact rational arithmetic. For a precision parameter P_0 , a constant multiple of P_0+1 terms in the series above suffices for P_0 output bits. Direct rational summation is enough: the products of the factorial and integer denominators have polynomial bit length. Square-root enclosures have polynomial cost in $P_0 + \log(c+1)$. Off-diagonal frequency denominators in (A.2) are nonzero integers times π , and $\pi > 3$.

Summing over at most B cells, multiplying weights bounded by c , and applying the row-sum estimate over $N_c < c$ entries lose only polynomial factors in B, c . Thus a common precision

$$P_0 = O(1 + \log(B+1) + \log(c+1) + \log(1/\tau)) \quad (\text{A.7})$$

suffices for (A.5) and the chosen eigenvalue width. Algorithmically, double P_0 until the explicit outward inequalities hold. The stated bound proves that this refinement terminates within polynomial bit cost. The bisection shifts also have polynomial bit length, so the minor bound applies at every inertia call. With $c = O(B^4 m^3 \tau^{-2})$, this proves that (A.6) is computable using a number of bit operations polynomial in B, m, τ^{-1} .

This is a numerical-parameter bound in the ordinary binary model: for $\tau = 2^{-s}$ it is not asserted to be polynomial in s . Multiplying this per-mask polynomial by the finite catalogue size leaves the outer exponential enumeration orders unchanged.

A.3. Effective certificates for a rational triangular mesh. The input now consists of a nondegenerate rational-vertex square, a finite conforming triangulation of its closure with rational vertices, a nonempty selection of triangles, an integer $m \geq 1$, and a positive rational τ . We show that rational intervals of width at most τ for $\lambda_1(P), \dots, \lambda_m(P)$ are computable. The proof supplies explicit mesh-quality certificates and entrywise errors, including all frequency coincidences.

Input validity is decidable by finite rational tests. Determinants check orientation and nondegeneracy; the square side vectors must have equal positive squared lengths and zero dot product. Check that all triangles lie in the closed square, that pairwise intersections are empty or a whole common edge or a common vertex, and that their areas sum to the square area. These conditions imply a conforming cover. Indeed, a missing interior point of their finite closed union would have a neighborhood of positive area; boundary coverage follows by closure.

At a triangle vertex let e, f be the outgoing edge vectors and $\Delta = |\det(e, f)| > 0$. The squared sine of its angle is the positive rational number $\Delta^2/(|e|^2|f|^2)$. Successive halving finds positive dyadic rationals θ, σ such that

$$\theta \leq 1, \quad \theta^2 \leq \min_{e,f} \frac{\Delta^2}{|e|^2|f|^2}, \quad \sigma^2 \leq \min_{T,e} \frac{\Delta_T^2}{|e|^2}. \quad (\text{A.8})$$

The last expression is the squared altitude to e . All minima include empty triangles. Since $\sin \alpha \geq \theta$ and $\sin \alpha \leq \alpha$, every angle is at least θ radians. No inverse trigonometric computation is needed.

Set the rational majorants

$$\overline{A} = \frac{4}{\theta}, \quad \overline{B} = \frac{4}{\theta^2}, \quad \overline{D} = 4\{24\overline{A}(1 + \overline{B}) + 2\}. \quad (\text{A.9})$$

The inequality $\sin x \geq 2x/\pi$ on $[0, \pi/2]$, together with $\pi < 4$ and $\theta \leq 1$, gives $D_\theta \leq \bar{D}$. An automatic rational ceiling for the first m energies is

$$E = 270m\sigma^{-2}. \quad (\text{A.10})$$

Indeed, the centroid of one occupied triangle is at distance at least $\sigma/3$ from each supporting edge. The coordinate-aligned square of side $\sigma/3$ centered there lies strictly in the triangle. It has rational vertices. Scaling the square estimate (13.23) gives $\lambda_m(P) < 270m\sigma^{-2}$.

Write the possibly rotated square as

$$Q = \{o + tv + uw : 0 < t, u < 1\}, \quad v \cdot w = 0, \quad |v|^2 = |w|^2 = L^2 > 0,$$

where o, v, w are rational. Every mesh vertex has rational (t, u) coordinates, L^2 is rational, and the absolute Jacobian is L^2 . The ambient eigenfunctions are $\phi_{kl} = 2L^{-1} \sin(k\pi t) \sin(l\pi u)$. Choose

$$c = \max \left\{ 3, \left\lceil \frac{4L^2 \bar{D}^2 (E + \sigma^{-2})^3}{9\tau^2} \right\rceil \right\}, \quad H = \frac{\pi^2 c}{L^2}. \quad (\text{A.11})$$

By (13.28) and $\pi^2 > 9$, the continuum error is at most $\tau/2$. The active list $\mathcal{I}_c = \{(k, l) \in \mathbb{N}^2 : k^2 + l^2 < c\}$ is obtained by integer comparisons and has size $1 \leq N_c \leq \pi c/4 < c$. Put

$$r_a = k_a^2 + l_a^2, \quad d_a = \sqrt{c - r_a}, \quad q = \pi^2/L^2, \quad G_{ab} = \int_P \phi_a \phi_b, \quad M_{ab} = q d_a d_b G_{ab}.$$

Then M is the physical Gram correction matrix M_H and $\ell_j(H) = H - \nu_j(M)$, with zero continuation beyond N_c . Although L need not be rational, its two normalization factors cancel the area Jacobian in each Gram entry.

A.4. Triangle integration without singular denominators. In the normalized (t, u) coordinates the integrand of G_{ab} is $4 \sin(k\pi t) \sin(k'\pi t) \sin(l\pi u) \sin(l'\pi u)$. For a normalized triangle T of rational area A and rational vertices x_0, x_1, x_2 , consider an affine phase $z(t, u) = at + bu$ with integers a, b , and put $z_i = z(x_i)$. The barycentric moment formula is

$$\int_T \lambda_0^i \lambda_1^j \lambda_2^k = \frac{2A i!j!k!}{(i+j+k+2)!}.$$

It follows by the affine change from the standard simplex and successive one-dimensional beta integrals; for integer powers those integrals are obtained by repeated integration by parts. Define the rational polynomial $h_r(z_0, z_1, z_2) = \sum_{i+j+k=r} z_0^i z_1^j z_2^k$. Expanding the affine phase and its exponential gives the absolutely convergent identity

$$\int_T e^{i\pi z(t,u)} dt du = 2A \sum_{r=0}^{\infty} \frac{(i\pi)^r}{(r+2)!} h_r(z_0, z_1, z_2). \quad (\text{A.12})$$

In particular, no differences $z_i - z_j$ occur in a denominator. Repeated phases, zero frequencies, and phases constant along an edge are covered by the same formula.

For real y , the integral Taylor remainder gives $|e^{iy} - \sum_{r=0}^K (iy)^r/r!| \leq |y|^{K+1}/(K+1)!$. An affine phase stays between its vertex extrema. With $Z = \max_i |z_i|$, the error in truncating (A.12) at K is therefore at most

$$A \frac{(4Z)^{K+1}}{(K+1)!}, \quad (\text{A.13})$$

using $\pi < 4$. The same estimate holds for the real part $C_T(a, b) = \int_T \cos(\pi(at + bu))$.

Set $a_- = k - k'$, $a_+ = k + k'$, $b_- = l - l'$, $b_+ = l + l'$, and $\epsilon_- = 1, \epsilon_+ = -1$. Twice using product-to-sum gives the triangle contribution

$$G_{ab}^{(T)} = \frac{1}{2} \sum_{\rho, \sigma \in \{-, +\}} \epsilon_\rho \epsilon_\sigma \{C_T(a_\rho, b_\sigma) + C_T(a_\rho, -b_\sigma)\}. \quad (\text{A.14})$$

The total absolute coefficient of these eight terms is four. For active modes and normalized vertices in $[0, 1]^2$, $|a_\rho| + |b_\sigma| \leq 4\sqrt{c} \leq 4c$. The selected normalized area is at most one. Hence the polynomial $P_{ab,K}(\pi)$ obtained by truncation satisfies the uniform rational bound

$$|G_{ab} - P_{ab,K}(\pi)| \leq 4 \frac{(16c)^{K+1}}{(K+1)!}. \quad (\text{A.15})$$

This tends to zero effectively. To obtain error at most η , increase K until the right side is at most $\eta/2$, then use (A.3) and rational interval polynomial evaluation until the value interval has width at most η . Its midpoint approximates G_{ab} within η . For a fixed polynomial the interval width tends to zero as the input interval shrinks, so both refinements terminate.

A.5. Rational matrix error and multiple eigenvalues. An explicit common tolerance can be chosen before the approximations. Put $Q_0 = 16/L^2$, so $0 < q < Q_0$, and set

$$\eta = \min \left\{ 1, \frac{\tau}{12c}, \frac{\tau}{48N_c(Q_0+1)(c+1)^2} \right\}. \quad (\text{A.16})$$

Compute rational approximations $\hat{q}, \hat{d}_a, \hat{G}_{ab}$ with absolute errors at most η . For the square roots, bisection of $x^2 - (c - r_a)$ on $[0, c]$ suffices. Compute one Gram approximation for each unordered pair and reflect it, preserving exact symmetry. Define $h = c\hat{q}$ and $\widehat{M}_{ab} = \hat{q}\hat{d}_a\hat{d}_b\hat{G}_{ab}$. Then $|H - h| \leq \tau/12$. Since $|G_{ab}| \leq 1$, $d_a \leq c$, $|\hat{d}_a| \leq c+1$ and $|\hat{q}| \leq Q_0+1$, a four-term telescoping expansion gives

$$|M_{ab} - \widehat{M}_{ab}| \leq \eta \{c^2 + (Q_0+1)c + (Q_0+1)(c+1) + (Q_0+1)(c+1)^2\} \leq \frac{\tau}{12N_c}.$$

Therefore

$$\|M - \widehat{M}\|_{\text{op}} \leq \|M - \widehat{M}\|_F \leq N_c \max_{a,b} |M_{ab} - \widehat{M}_{ab}| \leq \tau/12. \quad (\text{A.17})$$

The approximate matrix need not be positive. Symmetry and the norm error suffice for ordered eigenvalue comparison.

For a fully elementary terminating procedure, compute the characteristic polynomial of $\widehat{M}$ by exact rational arithmetic, perform its square-free factorization, and use Sturm isolation for its real roots. Symmetry ensures that all roots are real. Refine each distinct-root interval to width at most $\tau/6$; its midpoint has error at most $\tau/12$. Roots at subdivision endpoints are detected by exact substitution and treated as exact roots. The multiplicities recovered from square-free factorization give the ordered list, without an a-priori separation assumption. The exact inertia procedure above is an alternative.

Let ρ_j be the resulting midpoint for the j th decreasing eigenvalue of $\widehat{M}$, repeated with multiplicity, and continue $\rho_j = 0$ for $j > N_c$. By (A.17), $|\rho_j - \nu_j(M)| \leq \tau/6$ for $j \leq N_c$; beyond N_c both continued corrections are zero. Thus $a_j = h - \rho_j$ satisfies $|a_j - \ell_j(H)| \leq \tau/4$. Adding the continuum allowance proves

$$\lambda_j(P) \in [a_j - \tau/4, a_j + 3\tau/4], \quad 1 \leq j \leq m. \quad (\text{A.18})$$

These intervals have width τ and handle repeated levels.

The construction is finite and every numerical assertion is checkable by rational operations: one may retain the input geometry tests, θ, σ , the cutoff inequality, Taylor orders and remainders, scalar enclosures, the symmetric rational matrix, and the root counts. Its matrix dimension is

$$O(1 + L^2 \overline{D}^2 (E + \sigma^{-2})^3 \tau^{-2}),$$

or $O(1 + L^2 \overline{D}^2 m^3 \sigma^{-6} \tau^{-2})$ with (A.10). The global Taylor expansion was chosen to establish effectivity uniformly through phase degeneracies. No polynomial bit bound for this triangular implementation is needed or asserted.

A.6. Complete-basis checks and optional conforming upper bounds. Exact inertia proves bounded per-mask computation. For independently checking numerical proposals, a complete-basis residual test is also useful. Let A be rational symmetric, Q_0 a rational square matrix of proposed eigenvectors, and D_0 a diagonal matrix of ordered rational proposals. Suppose exact rational norm bounds verify

$$g \geq \|Q_0^T Q_0 - I\|_{\text{op}}, \quad g < \frac{1}{2}, \quad r \geq \|AQ_0 - Q_0 D_0\|_{\text{op}}, \quad d = \|D_0\|_{\text{op}}.$$

Frobenius bounds may be used throughout. For a symmetric matrix, the maximum absolute row sum also bounds its Euclidean operator norm. For the generally nonsymmetric residual $R = AQ_0 - Q_0 D_0$, one may instead use $\sqrt{\|R\|_1 \|R\|_\infty}$, where these are the maximum absolute column and row sums. With $G = Q_0^T Q_0$ and $K = Q_0^T A Q_0$, $G \geq (1 - g)I > 0$, so Q_0 is invertible, and

$$\|K - D_0\|_{\text{op}} \leq (1 + g)r + gd.$$

This follows by expanding $K - D_0 = Q_0^T (AQ_0 - Q_0 D_0) + (G - I)D_0$ and using $\|Q_0\| \leq \sqrt{1 + g} \leq 1 + g$. For any nonzero coefficient vector, comparing its generalized Rayleigh quotient for (K, G) with that for (D_0, I) adds at most another gd in the numerator and divides by $1 - g$. Min-max gives, at every ordered index,

$$|\lambda_j(A) - (D_0)_{jj}| \leq \frac{(1 + g)r + 2gd}{1 - g} =: \eta_A. \quad (\text{A.19})$$

The same statement applies after reversing both ordered lists. It certifies indices and multiplicities, not merely a nearby eigenvalue for each vector.

If A approximates T_c with norm error Δ , put $E_A = \Delta + \eta_A$. A proposal plus E_A is an upper endpoint for the corresponding decreasing correction eigenvalue and therefore gives a lower endpoint in (13.6). The endpoint's possible overhang is $2E_A$, not E_A : the proposal itself may already be E_A above the true value. For an outward scalar interval $[f_-, f_+]$ and dyadic output denominator S_0 , a sufficient total raw lower-endpoint arithmetic allowance is

$$2f_+ E_A + c(f_+ - f_-) + \frac{f_+}{S_0} + \frac{1}{S_0}. \quad (\text{A.20})$$

The last two terms pay for rounding the correction coordinate and the final product. Clamping the correction endpoint to $[0, c]$ and the final lower endpoint to $[0, \infty)$ preserves validity.

For upper bounds, subdivide the mask cells conformingly and use continuous piecewise affine functions with zero boundary nodal values. These are in the actual $H_0^1(P)$, including at point-contact vertices. The dyadic hierarchy is nested and dense: first approximate by a compactly supported smooth function and then interpolate on a sufficiently fine mesh. Its full Ritz values decrease to $\lambda_j(P)$. On a rectangle with side scales s_x, s_y divided into n intervals in each direction and split into right triangles, the local stiffness and consistent mass entries are rational:

$$K_{ij} = \frac{1}{2} \left(\frac{s_y}{s_x} b_i b_j + \frac{s_x}{s_y} c_i c_j \right), \quad M_{ij} = \frac{s_x s_y}{24n^2} \begin{cases} 2, & i = j, \\ 1, & i \neq j. \end{cases}$$

Here b_i, c_i are the signed edge differences in the reference unit right triangle. Assembly uses the consistent L^2 mass, which is positive definite on the free nodal space.

More generally, for a rational full-column-rank trial matrix C form $G = C^T M C$ and $K = C^T K_{\text{mesh}} C$ exactly. If $g \geq \|G - I\|_{\text{op}}$, $g < 1/2$, and $e \geq \|K - D_0\|_{\text{op}}$, min-max gives, at indices where $d_j - e > 0$,

$$\frac{d_j - e}{1 + g} \leq \text{Ritz}_j(C) \leq \frac{d_j + e}{1 - g}. \quad (\text{A.21})$$

The upper endpoint bounds the continuum eigenvalue. The lower endpoint bounds only the trial Ritz value and is not used as a continuum lower bound. A certified subspace may improve a finite interval without being a complete eigenbasis of the full finite-element matrix.

NOTE ON AI USE

The author declares the use of consumer-grade Gemini-3 and GPT-6 models for prior-art search, exploring and developing ideas and proofs, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

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Email address: arun.chandru@gmail.com